TEACHING DOSSIER

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1. My Teaching Philosophy

From my experience teaching at both universities such as Cornell University, the University of Toronto, Virginia Commonwealth University and The University of British Columbia and secondary schools such as Ithaca High school, I have drawn the following conclusions:

(1) People learn more in active learning classes than in traditional lectures.
(2) People learn best in an enviroment where they feel safe to make mistakes and ask questions.
(3) Focusing on reasoning rather than computation leads to deeper conceptual understanding.

I want my students to develop real mathematical skills and the ability to construct, analyze and critique arguments. I believe the best way of achieving this is to guide students to their own discoveries.

1.1. **Active learning.** I started as a teaching assistant during the first years of graduate school. I did not have any prior teaching experience nor any formal training. At the time, I thought that good teaching was just standing in front of students and presenting the material without making mistakes.

After teaching for a couple semesters, it became clear to me that this approach did not work for the majority of my students. I presented correct and careful solutions to standard problems during the tutorials, and I saw how the students who were nodding along could not solve very similar problems on the exams.

I wanted to become a more effective teacher and I started attending the Teaching Seminar at Cornell. One of the topics that we read papers on and discussed was active learning and I came to believe that people don't become experts by watching someone else perform a task. There is solid evidence that *active learning* is more effective than traditional lecture formats [FEM+14].

I have observed that giving students opportunity to express their ideas during class and in office hours allows them to gain independence and confidence in approaching and solving problems. Letting students explore also helps them develop a *growth mindset*.

In active learning environments, students learn more, but often they feel like they learn less [DMM+19]. For this reason, I like to take time to explain my methodology and emphasize the importance of why we go through certain steps rather than just finding the answer.

In my classes, most of the time is spent on students working on problems. I break down questions into small parts so that students can get started and make gradual progress. Students then share their ideas with their peers in small groups or a whole class discussion. I constantly communicate with the class to make sure that everyone is focused on the current task and is following the discussion.

The following comment from my recent course evaluations highlights my commitment to active learning (parts omitted):

"Balazs is probably the most inspiring instructor I have ever encountered . . . I would say the classroom culture of this class is unlike anything I have really experienced, I am quite introverted and was not keen to attend office hours (still am not), but particularly for this class I make the exception. Most instructors focus purely on learning from the textbook and having their students mindlessly regurgitate what they had read, but in this class we are asked to question and argue everything we read. In our piazza/classroom we have students not accepting what is taught to them and I feel like that is one of the biggest indicators that students are actively learning. This is probably the hardest class I have taken to date, but it is also the most I have ever been engaged. . . ."

1.2. **Creating a safe learning environment.** During my career, I have been able to maintain an atmosphere where students feel safe making mistakes and asking questions. The following comments from my recent course evaluations mean a lot to me:

- *"Very genuine and understanding, showed lots of care and put lots of time into helping. Also put lots of time and effort into the piazza which I really appreciated."*
- *". . . I've never asked questions during the classes before this class. I normally just waited until office hour and asked. However, he changed my attitude during the class and made me ask questions during class without any fear of feeling stupid and got to participate more. . . ."*
- *"I also really liked how patient and kind you were while explaining and answering questions. It made for a very supportive learning environment. Most math professors fall short of what is required not because of a lack of knowledge of their subject material, or because of an inability to articulate an explanation, but because the manner in which they respond to students is impatient. Making people feel stupid is hardly conducive for their learning. It was remarkable that you never once became impatient. . . "*
- *". . . He is really invested in ensuring that his students can understand the material. He is willing to meet with you outside of class regardless of how simple your question and regardless of your performance of the course. However, more importantly, he really and truly cares about the well-being of his students and does not see us as machines who began his class with the same level of knowledge. Professor Balazs understands that his students are human and that the things that are happening around them and at the colleges and dorms/homes can truly affect them. . . ."*

There is a large amount of research suggesting that *productive failure* and *productive struggle* can lead to more effective and deeper learning ([KB12] [Met17]).

There is often a perception that mathematics is overcompetitive and unwelcoming, where people are made to feel stupid for not being quick or lacking some knowledge. I have found that many students, especially those from underrepresented groups, have already internalized aspects of this culture and do not feel that they belong in mathematics classes. As a result, they lack the confidence to participate and ask questions in class. I try to mitigate this by making an effort to build a more collaborative and supportive environment.

I emphasize the importance of listening to the explanations and opinions of others, and I set an example for this by not cutting students off, and incorporating their ideas whenever possible. I consistently acknowledge the difficulty of the concepts we are considering and I avoid using words like "trivial" or "clear". I emphasize the role of making mistakes in the learning process, and I try to remove the social cost of being wrong by occasionally soliciting "wrong solutions" from students.

I have also found that an anonymous online discussion board like Piazza or Ed discussions can be an effective tool to let those students participate who would otherwise not do so in person.

1.3. **Focusing on reasoning.** I believe that the skill to construct and analyze arguments is more important than computing quickly and proficiently. Many students identify mathematics with manipulation of symbols, and focus on the rules and techniques of computation. In first-year Calculus classes, incoming students are often proficient at computing difficult derivatives, but lack the conceptual understanding of what their computations mean.

I was very happy when I received this comment in a recent course evaluation:

> *"He always looked at the working out and found issues even though we moved on from certain questions; leaving no students behind and ensuring we properly understood the intuition behind questions."*

While teaching a linear algebra class developed by Jason Siefken at the University of Toronto, I learned that asking students to find a mistake in an argument can be really beneficial for their learning. To encourage this sort of inquiry, I like to write problems that are phrased as fictional students discussing a question. There will be mistakes in their arguments, and the goal is to find and correct these mistakes. I found that this is an effective way of highlighting and dispelling common misconceptions.

2. Remarks about online learning

I have substantial experience facilitating online learning. The COVID-19 pandemic hit while I was teaching a linear algebra class, and we had to switch to online instruction at very short notice. During the Summer of 2020, I taught the same linear algebra class, and helped to design the course to be fully online.

I believe that in our class, students were able to learn as well as they could have in person. We were able to keep the active learning format despite large (over 200 students) class sizes and the online instruction. We placed a lot of emphasis on student interactions. One example is that students were randomly assigned to groups and were given reading assignments every week where they had to annotate and discuss the book chapters. This led to more students reading and discussing the textbook before class than in the in-person format, even though this was always a formal requirement.

We communicated our expectations clearly and we were also able to maintain a consistent online presence outside class time to help students with various difficulties. Student

feedback suggested that they really appreciated this effort, and they felt that their concerns were addressed promptly.

Unfortunately as many other courses, we had to deal with a large amount of academic integrity issues. Tests and examinations were fully online, and the temptation for students to use unautorized aids was large. We designed problems that were less prone to copying, monitored known cheating websites and asked them to take down the questions from our exams. Overall, our exams ended up going better than we initially expected.

3. Remarks about generative AI in education

Since the launch of ChatGPT 3.5, there have been many questions regarding large language models (LLMs) in education. Current models are capable of solving most standard homework problems that would be assigned in undergraduate courses. In addition, there is legitimate concern about the effect of students using LLMs as a tutor [KHY+25]. Generative AI can also make mistakes while seeming confident and authoritative, which can confuse non-experts.

One method I found effective was to take a question with an incorrect response and ask my students to critique it. My aim was to simultaneously improve their critical thinking skills and foster a healthy sense of skepticism when faced with generative AI outputs.

4. Conclusion

I have had the opportunity to teach mathematics at many different levels and institutions, and while each course has required me to adapt to new students and circumstances, I make sure to remain approachable and open to students' prior experience and perception of mathemtatics. I take care to engage my students and encourage them to participate and understand the importance of the process of learning, not just for a particular math class, but in their personal development. I have made a point to learn new methods and course aids to include all students in my courses and I look forward to applying all that I have learned about teaching so far in my career to provide a rewarding and positive learning experience for my future students.

5. Teaching experience

5.1. **Instructional experience.** I have instructional experience in various Calculus and Linear algebra classes, Combinatorics and Complex Analysis at universities such as Cornell, the University of Toronto, Virginia Commonwealth University and The University of British Columbia.

- During the Summer of 2019 I was the sole instructor for MAT344 Introduction to Combinatorics at the University of Toronto (the course website is here), where I had to oversee many of the administrative aspects that come with managing a large class.
- In the Fall semester of 2022 I was the course coordinator for MATH1120 Calculus II at Cornell University. I coordinated a team of five other instructors across eight lecture sections.
- In the Winter term of 2024 at The University of British Columbia I taught MATH223 Honours Linear Algebra (the course website is here) for which I received the following congratulatory letter from the Dean of Science:

THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Science

Faculty of Science, Office of the Dean
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February 28, 2025

Balazs Elek
Department of Mathematics
University of British Columbia
Mathematics Building, Zone 2

balazse@math.ubc.ca

Dear Balazs,

I was thrilled to learn that you have received some of the highest Student Experience of Instruction scores in the Faculty of Science in 2024 Winter Term 1 for MATH_V 223. Congratulations! You should be proud of how positively your students responded to your teaching. I know that it takes substantial time and effort to create meaningful and effective educational experiences for our students. Today's learners are ambitious, confident, and collaborative, and are eager to connect with each other and their instructors. The student experience ratings and comments from your course make it clear you are helping students succeed in their studies. The best students from Canada and from around the world come to UBC because they know they will experience the best science education possible, which is made possible by dedicated instructors, including you.

I want to offer you my personal thanks for all that you do for our students.

With my best wishes,

Well done, Balazs!
-Mark

Mark MacLachlan
Dean *pro tem*, Faculty of Science

CC: Dr. Daniel Coombs, Head, Department of Mathematics

I also have substantial experience teaching online, in Summer 2020 I was an instructor for an active learning linear algebra class, held fully online, and in the Spring of 2022 I taught a Calculus I class and two sections of Multivariable Calculus fully online at Virginia Commonwealth University.

5.2. **Course design experience.** During my final years of graduate school (2017-2018), the Cornell Mathematics Department transitioned to an active learning approach to Calculus 1. I was an instructor for Calculus 1 during these terms, and I participated in many of the discussions surrounding this transformation. In Summer 2020, also participated in redesigning the standard Linear algebra class at the University of Toronto to be completely online, while maintaining the focus on active learning. In Spring 2021, I developed a large amount of material for the MATH1910 Calculus for Engineers class to make it an active learning oriented class. In 2026, I have been working on developing worksheets for active tutorials for MATH2011, MATH1131 and MATH1231 at The University of New South Wales.

5.3. **Outreach.** A highlight of my teaching experience at Cornell was a course I taught several times at Ithaca High School, known as the Senior Math Seminar. This is an outreach program run jointly by the school and Cornell where graduate students in the math department can design and teach a course on any topic of their choosing to an audience of interested seniors in the school. I chose to teach a course on finite reflection groups (notes for the course can be found here). Working independently on a course was a fantastic opportunity, since I could experience aspects of teaching that a graduate student is not necessarily exposed to, such as developing class plans and materials for an audience with unusual backgrounds, without necessarily having an existing appropriate text to draw from. As always, I put a lot of emphasis to making the lectures engaging and interactive. This made it possible for high school students to succesfully follow material that is not usually covered even in undergraduate math courses.

I also participated in smaller outreach projects organized at Cornell for children as young as 12 years old. One example topic we covered in a program organized at Cornell in partnership with the Johns Hopkins Center for Talented Youth was a very basic introduction to group theory and symmetries using line dancing and the mattress flipping problem. The unique feature of this day-long outrech event is that the parents of the children (ages 10-15) also participate in the activities, but in a different group. We received positive feedback about our module from both the parents and the children.

5.4. **Mentoring experience.** The Ithaca High School Senior Math Seminar also involves students doing a project with the goal of presenting their findings a few months later, and I mentored several groups each year I was one of the instructors. This involved coming up with project ideas and meeting the students regularly.

In Spring 2020, I lead a reading course on quiver representations for a fourth-year undergraduate student at the University of Toronto. During the semester, I met with the student weekly to discuss his progress, suggest exercises or further reading, and provide

support. My student found the course very inspiring and later went on to participate in research projects as an undergraduate at the University of Toronto before starting a PhD program in Mathematics at the University of California, San Diego.

6. SAMPLE TEACHING MATERIAL

The following is a homework set that I wrote for MATH223 Honours Linear Algebra at The University of British Columbia. This course is the first serious proof-based class that students in the Math Honours program encounter. Given that the students are highly motivated and interested, I designed many of the homework sets to go quite far beyond the material of a standard linear algebra class to show applications in other areas of mathematics and introduce the use of mathematical software.

MATH223 Homework 6
(due Sunday, Nov/3, 11:59pm)

1. (this is Exercise 14 in Axler 4) Suppose $p, q \in \mathcal{P}(\mathbb{C})$ are nonconstant polynomials with no zeros in common. Let $m = \deg p$ and $n = \deg q$. Use linear algebra as outlined below in (a)-(c) to prove that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that
$$rp + sq = 1.$$
   (a) (2 marks) Define $T : \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) \to \mathcal{P}_{m+n-1}(\mathbb{C})$ by
$$T(r, s) = rp + sq.$$
   Show that the linear map T is injective.
   (b) (1 mark) Show that the linear map T in (a) is surjective.
   (c) (1 mark) Use (b) to conclude that there exist $r \in \mathcal{P}_{n-1}(\mathbb{C})$ and $s \in \mathcal{P}_{m-1}(\mathbb{C})$ such that $rp + sq = 1$.

2. As in question 1, let $p(x) = a_0 + a_1x + \ldots + a_mx^m$ and $q(x) = b_0 + b_1x + \ldots + b_nx^n$ be nonconstant polynomials (with $a_m \neq 0$, $b_n \neq 0$), but in this question do not assume that they do not share a root. Let $T : \mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C}) \to \mathcal{P}_{m+n-1}(\mathbb{C})$ be the same linear map
$$T(r, s) = rp + sq.$$
   (a) (1 mark) In question 1 you showed that T is an isomorphism if p and q do not share a root. Show the converse, i.e. prove that if T is an isomorphism then p and q do not share a root.
   (b) (2 marks) Recall that the list
$$\left((1, 0), (x, 0), \ldots, (x^{n-1}, 0), (0, 1), (0, x), \ldots, (0, x^{m-1})\right)$$
   is a basis of $\mathcal{P}_{n-1}(\mathbb{C}) \times \mathcal{P}_{m-1}(\mathbb{C})$ and that
$$(1, x, x^2, \ldots, x^{m+n-1})$$
   is a basis of $\mathcal{P}_{m+n-1}(\mathbb{C})$. Compute the matrix of T with respect to these bases.
   (c) (2 marks) The expression $\det(\mathcal{M}(T))$ is called the **resultant** of the polynomials p and q. If $\det(\mathcal{M}(T)) = 0$, then what can you say about the roots of p and q? Justify your answer.

3. As in questions 1 and 2, let $p(x) = a_0 + a_1x + \ldots + a_mx^m$ be a degree m polynomial. Let $q(x) = \frac{d}{dx}(p(x))$ be the derivative of p. We say that $p(x)$ **has a double root at** λ if $p(x) = (x-\lambda)^2 s(x)$ for some polynomial s of degree $m-2$.

   (a) (1 mark) Prove that $p(x)$ has a double root at λ if and only if $p(\lambda) = q(\lambda) = 0$ (**Hint:** This is a Calculus question).

   (b) (3 marks) Let $m = 2$ (i.e. p is a quadratic polynomial). Use question 2 to find a condition on the coefficients of p that is equivalent to p having a double root (**Hint:** remember, $a_2 \neq 0$). Do you recognize this expression? (**Hint:** It is generally best to leave the computation of determinants to a computer, especially if the matrix involved is large. There are numerous software that can compute determinants involving formal variables, sage is one of them that is relatively easy to use.)

   (c) (2 marks) Let $m = 3$ (i.e. p is a cubic polynomial). Find a condition on the coefficients of p that is equivalent to p having a double root.

4. In this question we'll explore another application of resultants. We want to find solutions of the system

$$\begin{aligned} y^2 &- (x^3 + 3x^2 + 2x + 1) = 0 \\ y^3 + y^2(3x+3) + y(3x^2+6x+2) &+ (x^3+3x^2+4x+2) = 0 \end{aligned}$$

   of non-linear polynomial equations. That is, we want to find pairs (x, y) of (real) numbers that satisfy both of the above equations.

   (a) (3 marks) Define two polynomials

$$\begin{aligned} p(x,y) &= y^2 - (x^3 + 3x^2 + 2x + 1) \\ q(x,y) &= y^3 + y^2(3x+3) + y(3x^2+6x+2) + (x^3+3x^2+4x+2). \end{aligned}$$

   Think of them as polynomials in the variable y (so, we treat functions of x as coefficients, notice that we already wrote them this way). Use question 2 to find a polynomial equation $s(x)$ in just the variable x that is equivalent to $p(x,y)$ and $q(x,y)$ sharing a root.[1]

   (b) (2 marks) Find a root of $s(x)$ and use this to find some solutions to the system[2] (you should be able to find three of them).

[1] This technique is one of the main tools of elimination theory. Notice how much harder polynomial elimination is than Gaussian (linear) elimination.

[2] This polynomial will be high degree so we don't have a formula for its roots. Computers can also help with factorization sometimes, here is a sage example.

6.1. **Sample lecture worksheet.** This is an example worksheet for a Calculus class where students are introduced the formula for the arc length of a curve. The formula seems quite unintuitive when first encountered, so my goal here was to lead students to gradually discover it. The idea of a polygonal approximation is introduced through a concrete example in part (b), and some of the difficulties of finding the length of the approximation are addressed in part (c). Once the students have seen some concrete computation, Exercise 2 leads them through a more abstract setting to the general formula.

MATH 1910 Lecture 16
Arc Length

Exercise 1. Consider the semicirle on Figure 1 Let's pretend that we don't know that the length

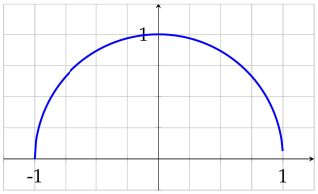


Figure 1: A semicircle

of this arc is π for this exercise.

(a) Write down a function $f(x)$ whose graph is exactly this semicircle.

(b) Let's approximate the length of this arc. Consider the approximation in Figure 2. Compute the length of the red curve.

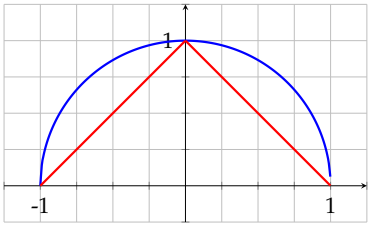


Figure 2: First approximation

(c) To improve the approximation, we increase the number of sample points. Compute the length of the red curve on Figure 3

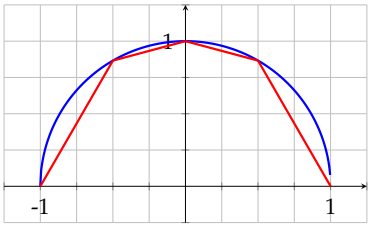


Figure 3: Second approximation

Exercise 2. Consider the approximating line segment on Figure 4. Assume that the blue curve is the graph of the function $f(x)$, and that the two blue circles are at $(x_i, f(x_i))$ and $(x_{i+1}, f(x_{i+1}))$, respectively. Let $\Delta x = x_{i+1} - x_i$ (the length of the green line segment on the figure).

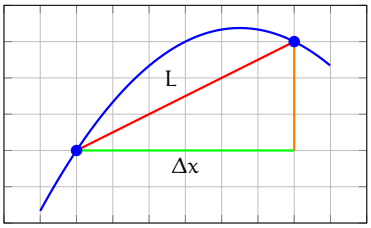


Figure 4: The length of a line segment

(a) How would you express the length of the orange segment?

(b) How can we express the length L of the red line segment?

(c) If f' exists and is continuous, then the Mean value theorem says that there is a value c_i in $[x_i, x_{i+1}]$ such that

$$f(x_{i+1}) - f(x_i) = f'(c_i)(x_{i+1} - x_i) = f'(c_i)\Delta x.$$

Substitute this into the formula for L, can you pull anything out from under the square root?

(d) Let s denote the arc length of $y = f(x)$ over $[a, b]$. Explain the two equalities below:

$$s = \lim_{n\to\infty} \sum_{i=1}^{n} \sqrt{1 + (f'(c_i))^2}\, \Delta x = \int_a^b \sqrt{1 + f'(x)^2}\, dx.$$

6.2. **Sample homework question.** This is a homework question for a Calculus for Engineers class in Spring 2021. The students were assigned regular WeBWorK assignments where they can develop computational proficiency, and I wanted to complement this with critique-style questions where students have to analyze a given argument and find the mistakes. This is supposed to simulate the in-class student discussions in an active learning classroom. My goal with this question is to highlight a common misunderstanding about manipulating equations involving indefinite integrals.

Problem 1. (8 points) Odysseus and Penelope are trying to compute the indefinite integral

$$\int \frac{1}{x\ln(x)}\,dx.$$

Having just learned integration by parts, they are keen to apply this method. They choose $u = \frac{1}{\ln(x)}$ and $dv = \frac{1}{x}dx$ and they proceed as follows:

(i) Penelope then computes $du = \frac{1}{(\ln(x))^2}\cdot\frac{1}{x}dx$, $v = \ln(x)$.

(ii) Odysseus then writes out the result of the integration by parts formula

$$\int \frac{1}{x\ln(x)}\,dx = \frac{\ln(x)}{\ln(x)} + \int \frac{\ln(x)}{(\ln(x))^2 x}\,dx$$

(iii) Penelope simplifies the righ-hand side to obtain

$$\int \frac{1}{x\ln(x)}\,dx = 1 + \int \frac{1}{\ln(x)x}\,dx$$

(iv) Odysseus then says the following:

We must have made a mistake. We got back the integral that we started with, so we can cancel $\int \frac{1}{x\ln(x)}\,dx$ *from both sides and then we get*

$$0 = 1.$$

(v) Penelope thinks that they must have made a sign mistake somewhere, it happens all the time with integration by parts, so the answer then is probably

$$\int \frac{1}{x\ln(x)}\,dx = 1 - \int \frac{1}{\ln(x)x}\,dx$$

and this would lead to

$$2\int \frac{1}{x\ln(x)}\,dx = 1,$$

therefore

$$\int \frac{1}{x\ln(x)}\,dx = \frac{1}{2},$$

(a) (4 points) Find all the mistakes in Odysseus and Penelope's argument. Make sure to explain in complete sentences any of the conceptual mistakes they made. In particular, what is wrong with the last equation they have? (**Hint:** it isn't just one sign mistake)

(b) (2 points) Find $\int \frac{1}{x\ln(x)}\,dx$. Check that you got the correct answer by differentiating. (**Hint:** you might want to try substitution instead of integration by parts.)

6.3. **More sample questions.** These are more homework quesitons from the Spring 2021 Calculus for Engineers class. During the final few weeks of the course, students are introduced the formal definition of a limit of a sequence. My goal with these questions was to highlight the importance of interpreting and using definitions very carefully. I designed the modifications to correspond to concepts that the students have encountered before: "the sequence is eventually constant" and "the sequence is bounded", and many of them successfully answered part (c) of these questions.

Problem 1. (6 points) Consider the following modification of the definition of a limit.

Definition 1. *We say that the **lumit** of a sequence* $\{a_n\}$ *is* L *and write*

$$\operatorname*{lum}_{n\to\infty} a_n = L$$

if there exists an $N > 0$ *such that for all* $\varepsilon > 0$*,*

$$|L - a_n| < \varepsilon$$

for all $n > N$*.*

For each of the following parts, give a proof or a concrete counterexample.

(a) (2 points) Assume that $\lim_{n\to\infty} a_n = L$. Is $\operatorname{lum}_{n\to\infty} a_n = L$?

(b) (2 points) Assume that $\operatorname{lum}_{n\to\infty} a_n = L$. Is $\lim_{n\to\infty} a_n = L$?

(c) (2 points) An informal way to describe a sequence $\{a_n\}$ with $\lim_{n\to\infty} = L$ is to say:

As n *increases, the values of* a_n *get arbitrarily close to* L*.*

How would you informally describe a sequence with $\operatorname{lum}_{n\to\infty} a_n = L$?

Problem 2. (6 points) Consider the following modification of the definition of a limit.

Definition 2. *We say that the* ***lamit*** *of a sequence* $\{a_n\}$ *is* L *and write*

$$\operatorname*{lam}_{n\to\infty} a_n = L$$

if there exists an $\varepsilon > 0$ *such that there exists* $N > 0$ *such that*

$$|L - a_n| < \varepsilon$$

for all $n > N$.

For each of the following parts, give a proof or a concrete counterexample.

(a) (2 points) Assume that $\lim_{n\to\infty} a_n = L$. Is $\operatorname{lam}_{n\to\infty} a_n = L$?

(b) (2 points) Assume that $\operatorname{lam}_{n\to\infty} a_n = L$. Is $\lim_{n\to\infty} a_n = L$?

(c) (2 points) An informal way to describe a sequence $\{a_n\}$ with $\lim_{n\to\infty} = L$ is to say:

As n *increases, the values of* a_n *get arbitrarily close to* L.

How would you informally describe a sequence with $\operatorname{lam}_{n\to\infty} a_n = L$?

6.4. **Sample outreach material.** The lecture notes are from the outreach course on Reflection groups at the Ithaca High School. The audience consists of highly motivated high school students of ages 16-18, but they do not have any background university-level mathematics. I tried to write in a language that they understand, and we did two hands-on examples thoroughly before we got to the definitions. In class, students would walk up to the smartboard and work out the reflections with color.

Exercise 3.7. *Identify the cross-polytopes in Figures 5, 6, and 8.*

Exercise 3.8. *What happens if you dualize the n-simplex?*

Exercise 3.9. *What happens if you dualize the dodecahedron? The icosahedron?*

Exercise 3.10. *Show that dual polytopes have identical symmetry groups.*

So far we have seen three infinite families of regular polytopes: the n-simplices, the n-cubes and their duals, the n-cross-polytopes. But how about the other guys? The dodecahedron, icosahedron, 24-cell, 120-cell and 600-cell? It turns out that it is dimension 4 where their story ends. They have absolutely no analogues in higher dimensions, moreover, from dimensions 5 onwards, the only regular polytopes are the n-simplices, the n-cubes and the n-cross-polytopes. The reason behind this fascinating result is the lack of exceptional finite reflection groups that would serve as these polyotope's symmetry groups.

4 Root systems

We have remarked previously that the reason for the lack of regular polytopes in higher dimensions was the lack of suitable symmetry groups. We are now going to classify finite reflection groups. The way we are going to do this is to find a nice combinatorial gadget that we can associate to a reflection group, and classify those.

4.1 A_2 and B_2

We have defined reflectiong groups to be groups generated by reflections in Euclidean space. However, looking at the examples in Section 2.4, it seems that all the stuff in $\mathbb{R}^n$ is not really necessary to describe our reflection group W. For instance, if we know s sends a vector $\mathbf{v}$ to $-\mathbf{v}$, then s sends every vector on the line through the origin and $\mathbf{v}$ to their negatives. It would be nice if we could think about what W does without concerning ourselves with all of $\mathbb{R}^n$. In particular, it would be great if we could find a *finite* set of vectors in $\mathbb{R}^n$, which are preserved by W, and are sufficient to describe its behavior.

Let's see what we can do with the symmetry group of the regular triangle (a.k.a. S_3, but later we'll call it A_2).

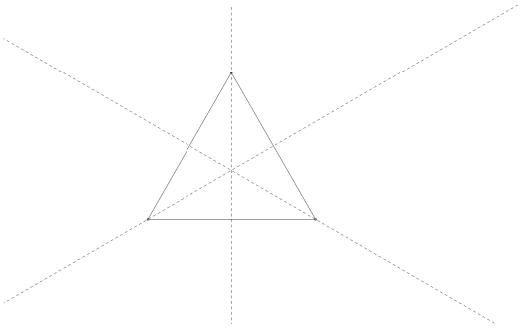


Figure 14: Symmetry lines of the regular triangle

Since any element of S_3 will preserve the triangle, it will also preserve these lines. However, we would like to keep track of the reflections sending certain vectors to their negatives, so let's pick two vectors of equal length that are perpendicular to each line

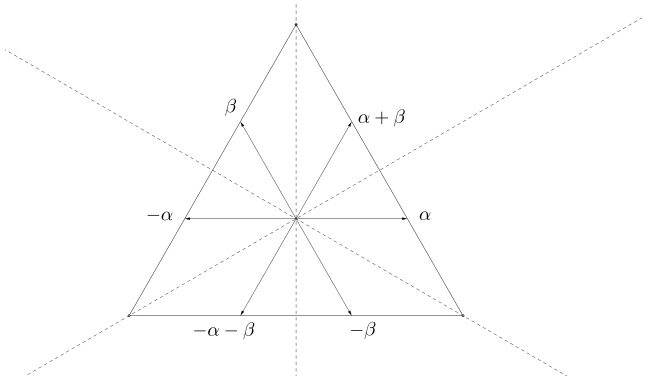


Figure 15: The root system of A_2

Now notice that any reflection along the symmetry lines will permute the set $\{\alpha, \beta, \alpha+\beta, -\alpha, -\beta, -\alpha-\beta\}$. So we have managed to turn this geometric action of a reflection group into combinatorics. We can write down in a table what happens to the roots if we apply elements of A_2 to them.

	α	β	$\alpha+\beta$	$-\alpha$	$-\beta$	$-\alpha-\beta$
e	α	β	$\alpha+\beta$	$-\alpha$	$-\beta$	$-\alpha-\beta$
s_α	$-\alpha$	$\alpha+\beta$	β	α	$-\beta-\alpha$	$-\beta$
s_β	$\alpha+\beta$	$-\beta$	α	$-\alpha-\beta$	β	$-\alpha$
$s_\alpha s_\beta$	β	$-\alpha-\beta$	$-\alpha$	$-\beta$	$\alpha+\beta$	α
$s_\beta s_\alpha$	$-\alpha-\beta$	α	$-\beta$	$\alpha+\beta$	$-\alpha$	β
$s_\alpha s_\beta s_\alpha$	$-\beta$	$-\alpha$	$-\alpha-\beta$	β	α	$\alpha+\beta$

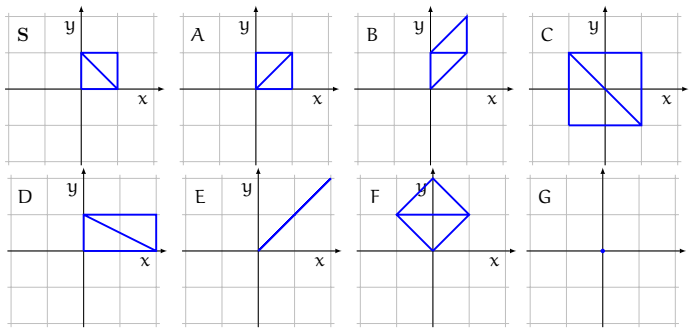
Table 1: The action of A_2 on its root system

Exercise 4.1. *If someone covers the first column of Table 1, how would you try to identify which rows correspond to reflections and which to rotations?*

Let's try to replicate this for the symmetry group of the square (which later will be known as B_2).

6.5. **An exam question.** This is a question I wrote for the final exam of the Summer 2020 class MAT223 Linear Algebra at the University of Toronto, which was held fully online, and we had significant trouble with cheating websites such as Chegg during the term. My goal with this question was twofold: to reduce or prevent cheating, and to focus on the conceptual understanding and reasoning rather than computation. I have received positive feedback from teaching experts concerning this question, and students also enjoyed solving it.

Out of pure interest a student uploads some of the pictures from our Quiz 2 to Chegg:



(note: some of the labels have changed from Quiz 2) and asked: Here S is a picture of a set and $A-G$ are images of the set under some transformation. For each of these transformations, find all possible eigenvalues and eigenvectors. Since Chegg pays their workers per answer, they are incentivized to answer as many questions as they can during a given time.

The exploited Chegg worker had the following quick thoughts about each picture before having to move on to solve another student's Calculus exam for them while wondering about how much profit Chegg will be making in this exam period:

(A) This looks like a rotation, so it shouldn't have real eigenvalues. Or maybe it's a reflection? I'll just say something like "rotations must have complex eigenvalues, so they can't have real ones". *This sounds very knowledgeable, and if the student is asking this question on Chegg they won't check what I say anyway.*

(B) This is a tilting, so nothing is being stretched here, no eigenvalues. *Just like the tilt that the instructors will feel when they see this question appear on Chegg.*

(C) Finally an easy one, clearly everything is scaled up by a factor of two, so everything is an eigenvector with eigenvalue 2.

(D) Any vector on the x-axis is an eigenvector with eigenvalue 2. *This means a stretch by a factor of two, just like the speed of the course whose final exam this is an actual question on.*

(E) This is a projection, so it's not invertible and therefore it can't have eigenvalues. *Just like you can't learn effectively by copying solutions someone else wrote.*

(F) We can see that the diagonal line in the original square has been scaled up by a factor of $\sqrt{2}$, so any vector on it is an eigenvector with eigenvalue $\sqrt{2}$.

(G) This transformation sends everything to zero. *Just like what happens to students' knowledge when they ask exam questions on Chegg.* Since only nonzero vectors can be eigenvectors, this does not have eigenvectors at all.

Point out **all** the mistakes in the Chegg worker's arguments using **at most two** sentences each (2 points for each picture). **Ignore** everything that is in italicized font.

7. TEACHING EVALUATIONS

Research shows that numerical student evaluation scores are correlated with factors such as gender, race and appearance, but not strongly correlated with actual learning. Therefore, I mostly focus on the student comments and their evaluations of the other aspects of course, such as approachability and organization.

I tend to get comments that students feel that I am a patient and approachable, and that they feel comfortable asking questions and generally participating in my classes. Students also feel that they get a lot out of my office hours.

One area that I would like to improve in is establishing student buy-in for active learning. Generally speaking students are very comfortable with traditional lecture formats and they often feel that any form of struggle during class is unproductive. I have some activities that I use to convince students that active learning is the best way for them to learn, but I would like to do more.

7.1. **Complete evaluation history.** Below is a selected list of my recent student evaluations, full evaluations are available upon request

UNSW SYDNEY | Australia's Global University

myExperience Report

Term 1, 2026

Faculty: Faculty of Science

School: Sch Mathematics & Statistics

Course: MATH1131 Mathematics 1A

Evaluation period: Apr 13 2026 12:00AM - Apr 30 2026 12:00AM

Teacher Report: Balazs Elek

Response Data for Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
Response Count	68

Comparison of results for "Overall I was satisfied with the quality of this person's teaching"

This teacher: Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Options	Count	Percentage
Strongly disagree	1	1.5%
Disagree	1	1.5%
Moderately disagree	0	0.0%
Moderately agree	3	4.4%
Agree	13	19.1%
Strongly agree	50	73.5%

Statistics	Value
Mean	5.59
Median	6.00
Standard Deviation	0.90
% Agree broad	97.1%

The table below shows the percentage of 'Agree' and 'Strongly agree' responses to the question 'Overall I was satisfied with the quality of this person's teaching'

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
% Agree	92.6%

SCHOOL: Sch Mathematics & Statistics

Overall I was satisfied with the quality of this person's teaching

Options	Percentage
Strongly disagree	0.9%
Disagree	1.1%
Moderately disagree	2.4%
Moderately agree	9.8%
Agree	35.2%
Strongly agree	50.5%

Statistics	Value
Mean	5.29
Median	6.00
Standard Deviation	0.93
% Agree broad	95.6%

FACULTY: Faculty of Science

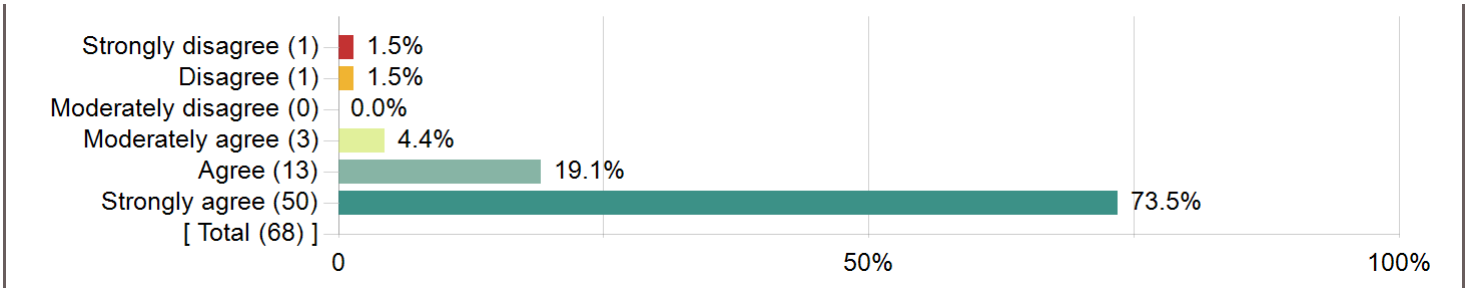
Overall I was satisfied with the quality of this person's teaching

Options	Percentage
Strongly disagree	1.0%
Disagree	1.1%
Moderately disagree	2.3%
Moderately agree	9.5%
Agree	34.8%
Strongly agree	51.3%

Statistics	Value
Mean	5.30
Median	6.00
Standard Deviation	0.93
Standard Error (based on SD)	0.01
% Agree broad	95.6%

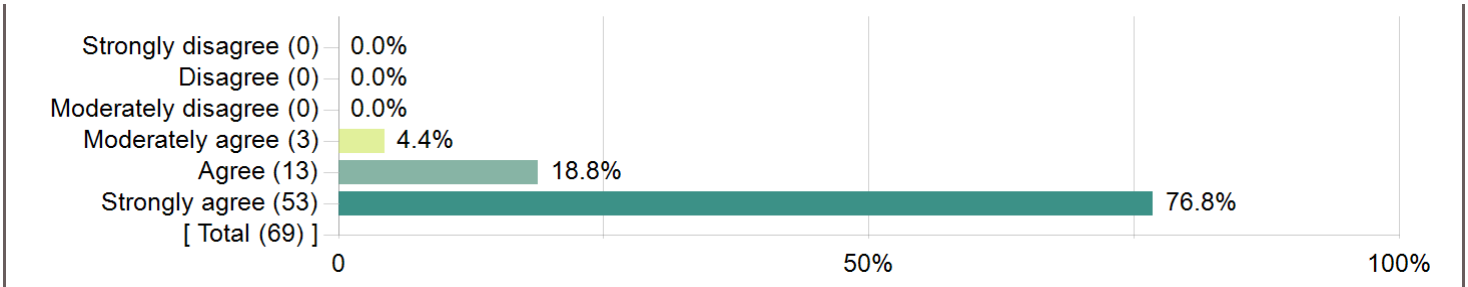
Overall I was satisfied with the quality of Balazs Elek's teaching

Overall I was satisfied with the quality of this person's teaching



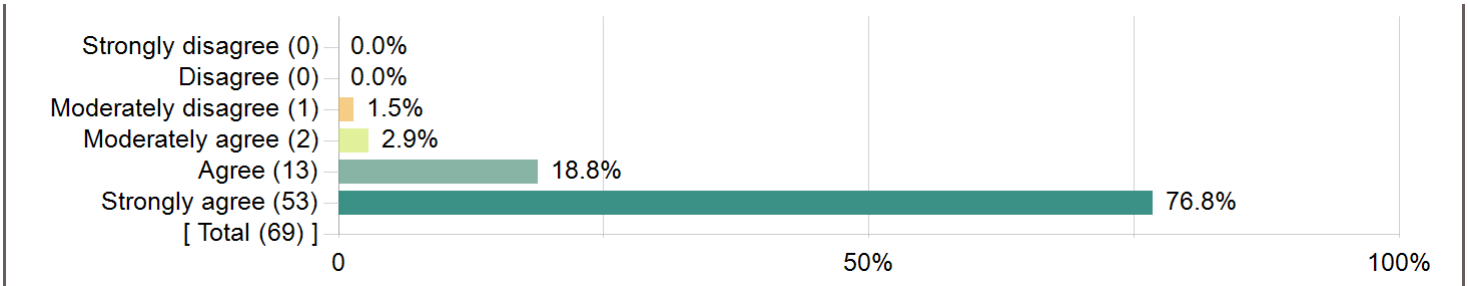
Statistics	Value
Mean	5.59
% Agree broad	97.1%

1. Balazs Elek encouraged student participation



Statistics	Value
Mean	5.72
% Agree broad	100.0%

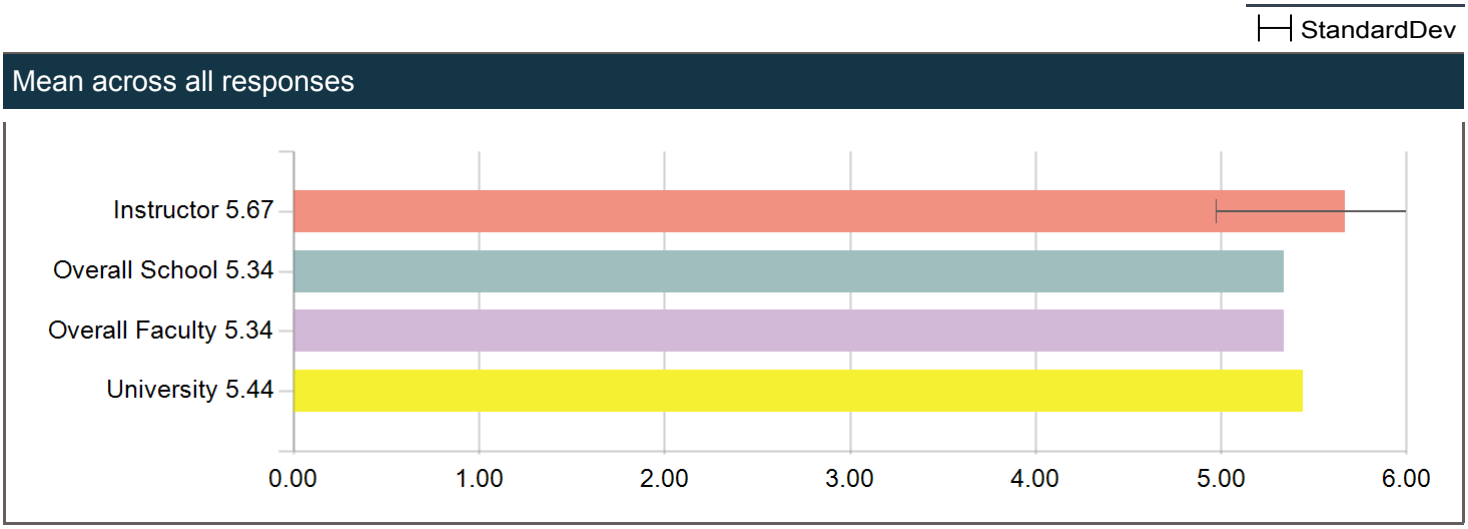
2. Balazs Elek provided helpful feedback



Statistics	Value
Mean	5.71
% Agree broad	98.6%

Comparison Statistics

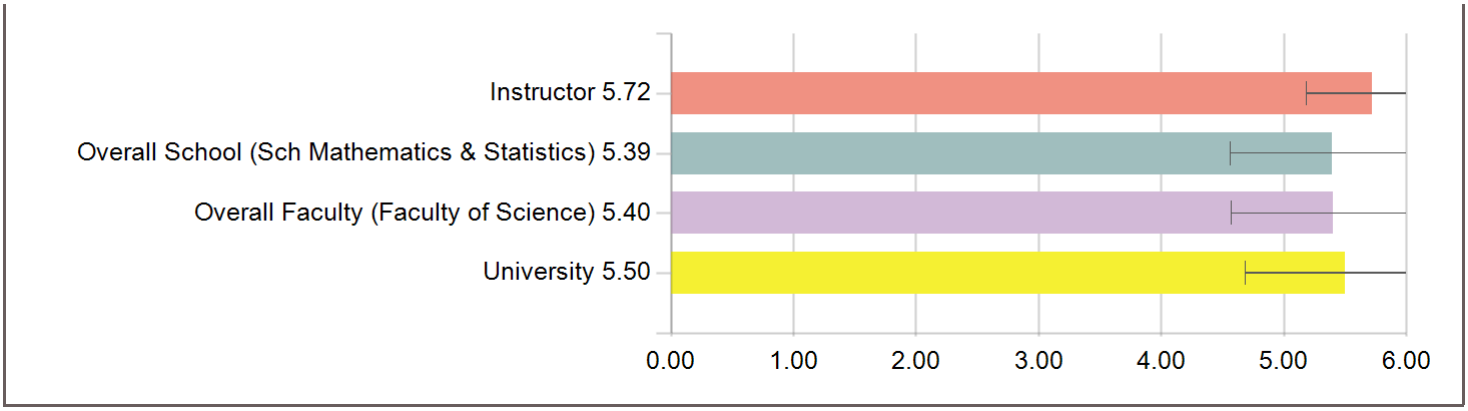
Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

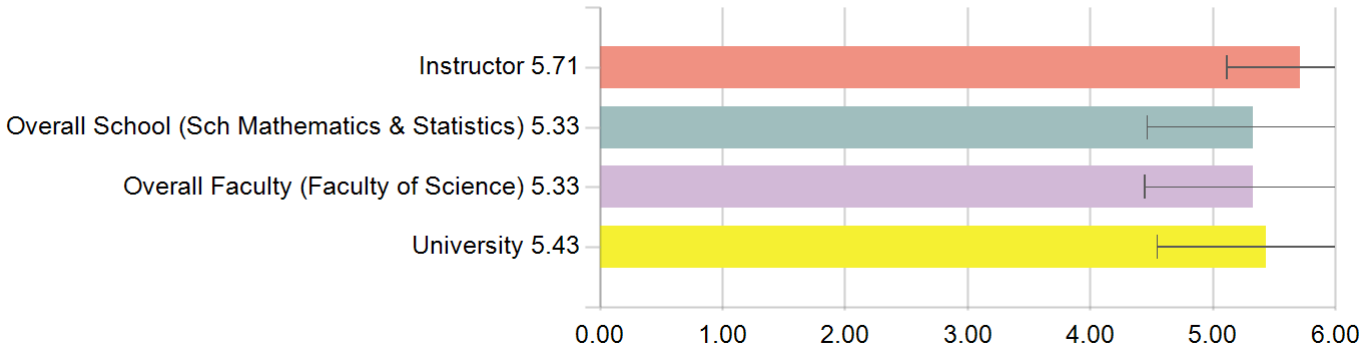
1. encouraged student participation



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

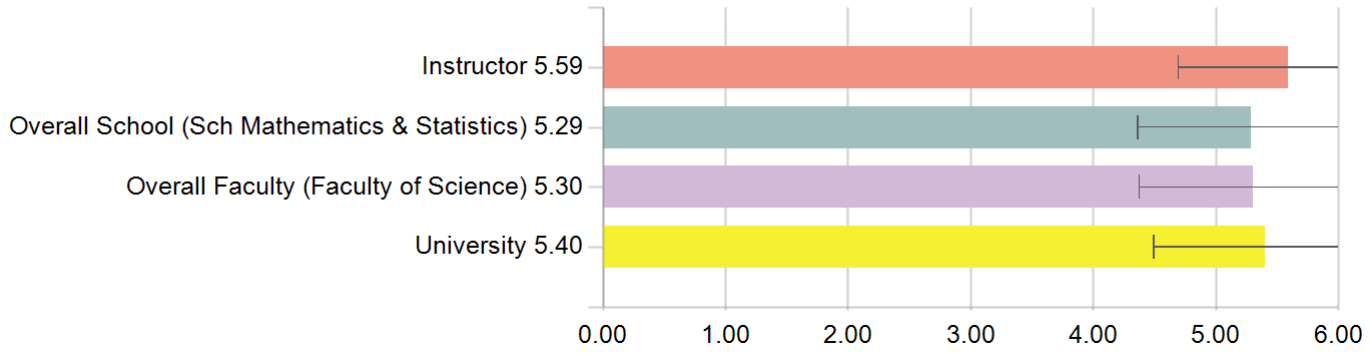
2. provided helpful feedback



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

3. Overall I was satisfied with the quality of this person's teaching



Raw comment data

The best features of Balazs Elek's teaching were

Comments
Very engaging, informative and explains questions until everyone understands
his explanations and the way he expresses it.
In depth analysis
The Goat
Positive attitude and calm teaching.
I think out tutor was extremely competent and was great with our class. He always asked us what questions we wanted to cover that lesson, ensuring that any content we were unsure of was discussed, and provided multiple ways of solving questions to deepen our understanding. He never rushed through questions, rather taking his time on topics to ensure understanding. He would always attempt to provide an answer, even when questions weren't specific to tutorials, such as mobius and gave valuable and helpful advice on exams and how to approach questions.
that he was the greatest of all time
He clearly and in depth taught ideas
Balaz was an engaging and wonderful tutor and I was so happy to have him for my tutorials. He was very good at breaking down problems into smaller bits and dissecting concepts we didn't know. He always came with a smiles and was truly wonderful to have teach us. Thanks Balaz!
He encouraged heaps of student participation and always took his time to explain questions in multiple ways. The class never felt rushed and it was all based off what the students felt like they needed help with rather than a preassigned schedule.
His clear explanations of complex concepts made it easy to learn new content
The class discussion of the problems was very detailed and very interesting.
It's very well explained. Every time there is a question that I don't understand, it will be resolved !

Balazs Elek's teaching could be improved by

Comments
a raise
no need for improvement
If Balaz could maybe check we know every step of what we are teaching that would be useful, checking in if we understand each step a bit more. This happened a few times but students often asked questions if they didnt understand anyways.

myExperience Report

Term 1, 2026

Faculty: Faculty of Science

School: Sch Mathematics & Statistics

Course: MATH1141 Higher Mathematics 1A

Evaluation period: Apr 13 2026 12:00AM - Apr 30 2026 12:00AM

Teacher Report: Balazs Elek

Response Data for Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
Response Count	29

Comparison of results for "Overall I was satisfied with the quality of this person's teaching"

This teacher: Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Options	Count	Percentage
Strongly disagree	0	0.0%
Disagree	0	0.0%
Moderately disagree	0	0.0%
Moderately agree	1	3.4%
Agree	11	37.9%
Strongly agree	17	58.6%

Statistics	Value
Mean	5.55
Median	6.00
Standard Deviation	0.57
% Agree broad	100.0%

The table below shows the percentage of 'Agree' and 'Strongly agree' responses to the question 'Overall I was satisfied with the quality of this person's teaching'

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
% Agree	96.6%

SCHOOL: Sch Mathematics & Statistics

Overall I was satisfied with the quality of this person's teaching

Options	Percentage
Strongly disagree	0.9%
Disagree	1.1%
Moderately disagree	2.4%
Moderately agree	9.8%
Agree	35.2%
Strongly agree	50.5%

Statistics	Value
Mean	5.29
Median	6.00
Standard Deviation	0.93
% Agree broad	95.6%

FACULTY: Faculty of Science

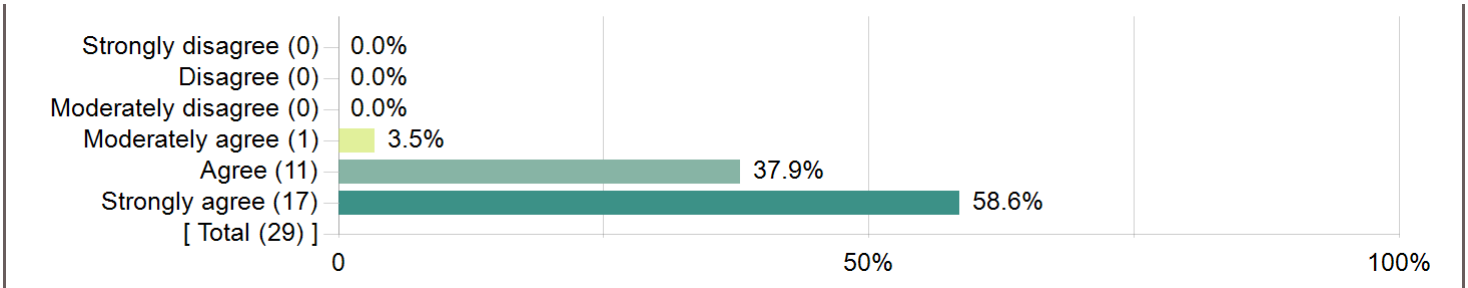
Overall I was satisfied with the quality of this person's teaching

Options	Percentage
Strongly disagree	1.0%
Disagree	1.1%
Moderately disagree	2.3%
Moderately agree	9.5%
Agree	34.8%
Strongly agree	51.3%

Statistics	Value
Mean	5.30
Median	6.00
Standard Deviation	0.93
Standard Error (based on SD)	0.01
% Agree broad	95.6%

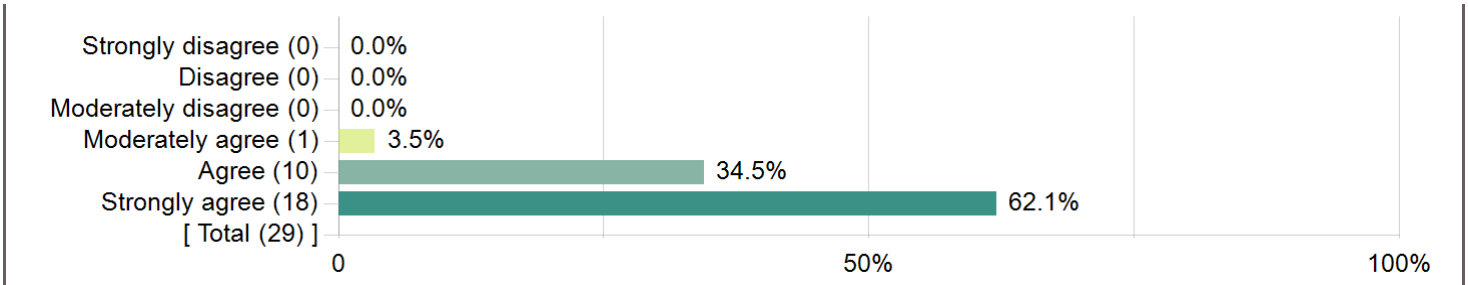
Overall I was satisfied with the quality of Balazs Elek's teaching

Overall I was satisfied with the quality of this person's teaching



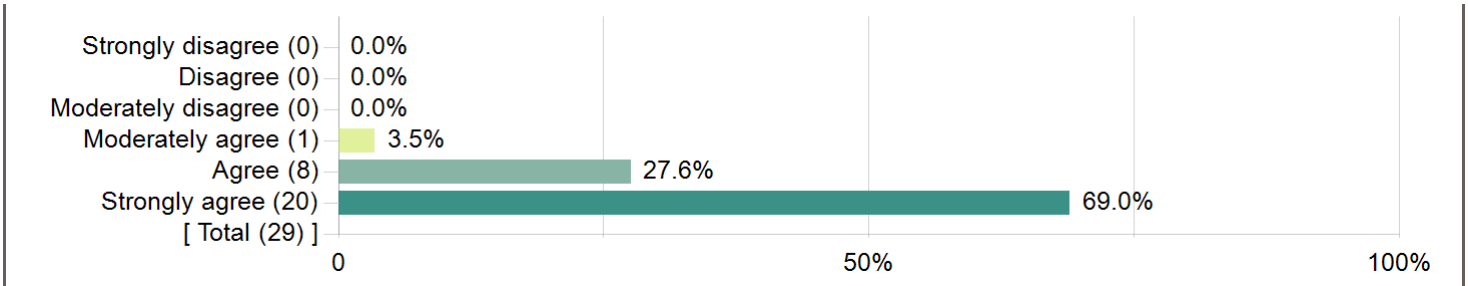
Statistics	Value
Mean	5.55
% Agree broad	100.0%

1. Balazs Elek encouraged student participation



Statistics	Value
Mean	5.59
% Agree broad	100.0%

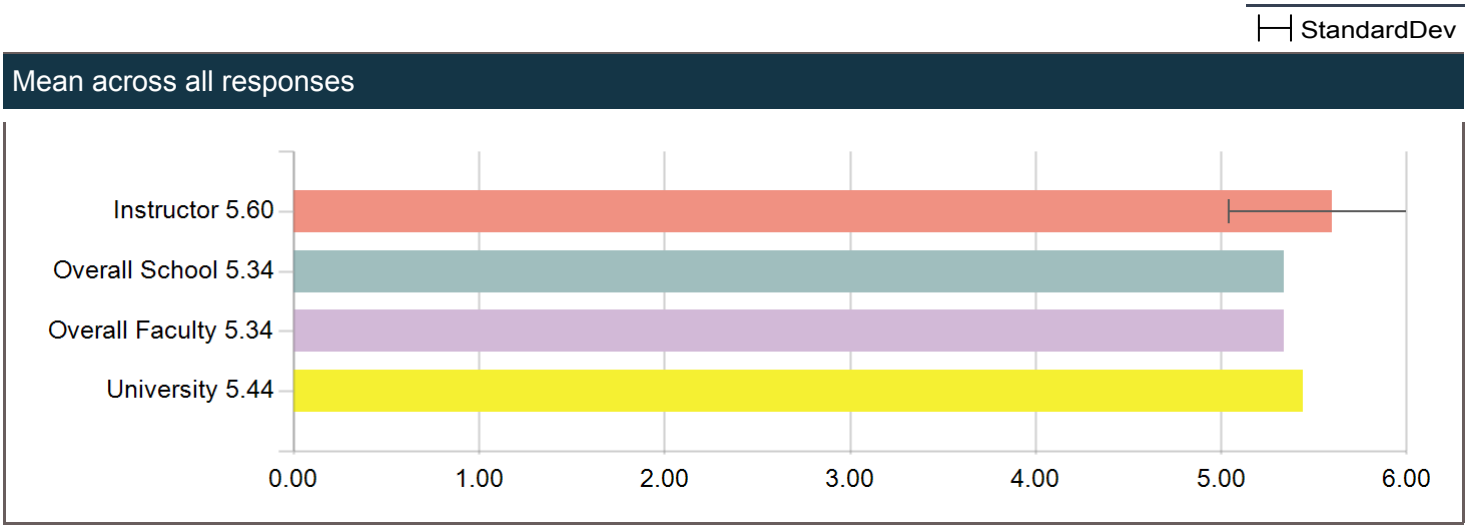
2. Balazs Elek provided helpful feedback



Statistics	Value
Mean	5.66
% Agree broad	100.0%

Comparison Statistics

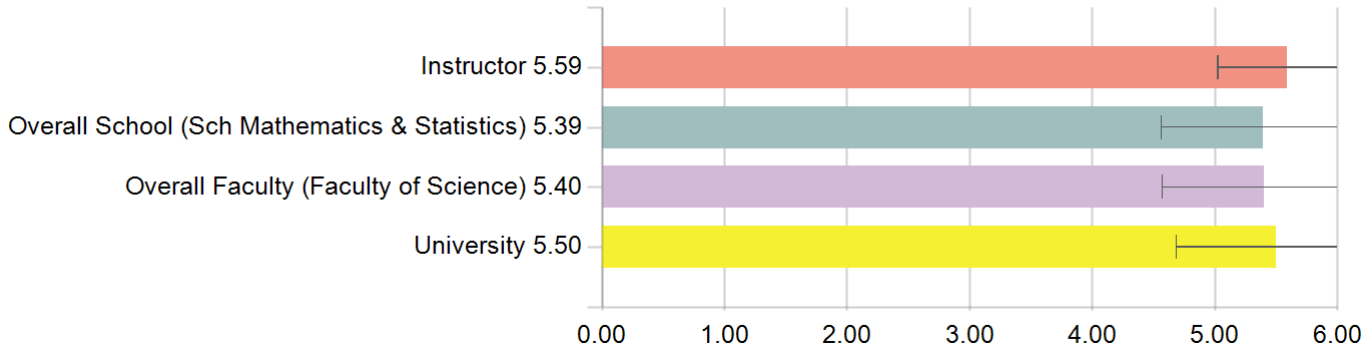
Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

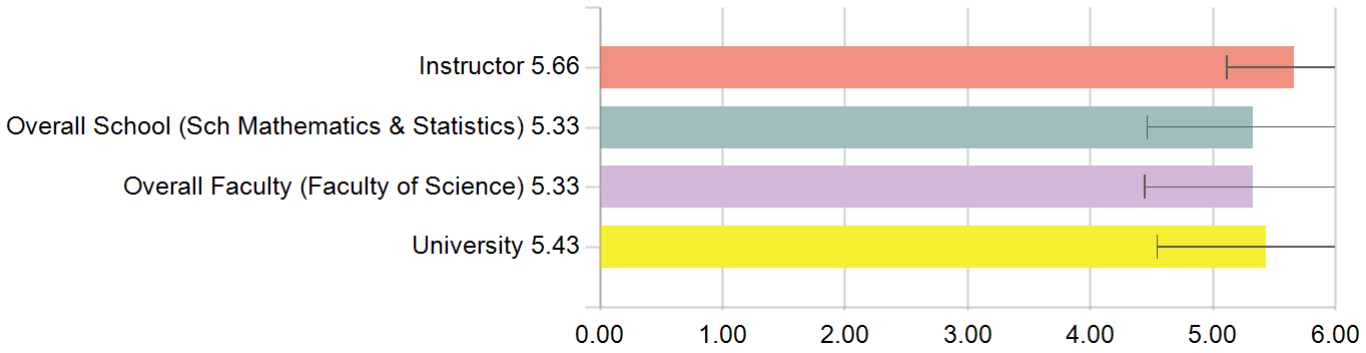
1. encouraged student participation



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

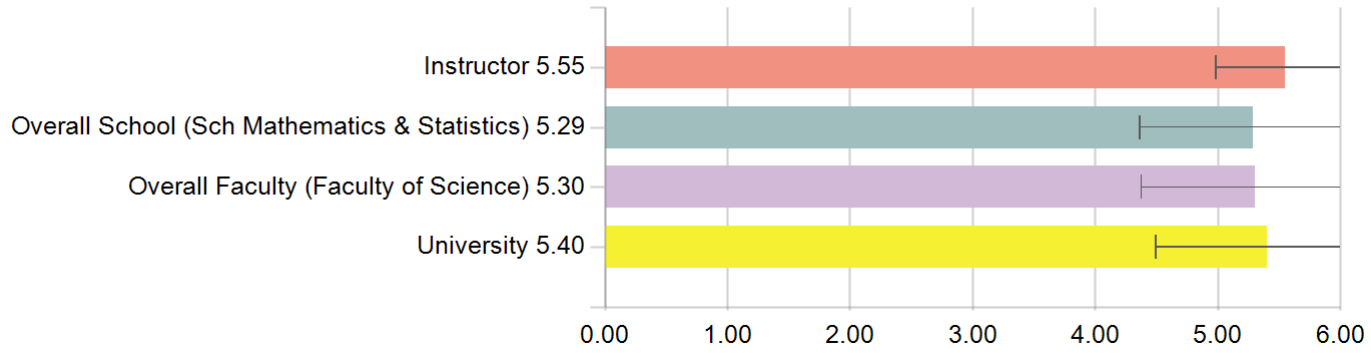
2. provided helpful feedback



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

3. Overall I was satisfied with the quality of this person's teaching



Raw comment data

The best features of Balazs Elek's teaching were

Comments
He encouraged us to develop intuition. I had genuinely good time whenever he helped us!
I think one of the best parts of his teaching method was knowing exactly when to jump in and help a student when they're stuck on a question. He was also really good at giving helpful hints and constructive feedback when I was stuck on a question, and when he did explain the answer, it was clear and concise. Thank you for teaching me this term!
Nice, helpful, patient, and encourage critical thinking.
Going around to each group and giving feedback on answers throughout the class time
He always looked at the working out and found issues even though we moved on from certain questions; leaving no students behind and ensuring we properly understood the intuition behind questions.
He was very good at pointing out flaws and suggesting possible corrections in our board tutorials.
He would always come to your group and explain a concept very well.
Explained concepts well

Balazs Elek's teaching could be improved by

Comments
N/A
Lessons are too short

myExperience Report

Term 1, 2026

Faculty: Faculty of Science

School: Sch Mathematics & Statistics

Course: MATH2011 Several Variable Calculus

Evaluation period: Apr 13 2026 12:00AM - Apr 30 2026 12:00AM

Teacher Report: Balazs Elek

Response Data for Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
Response Count	15

Comparison of results for "Overall I was satisfied with the quality of this person's teaching"

This teacher: Balazs Elek

Overall I was satisfied with the quality of this person's teaching

Options	Count	Percentage
Strongly disagree	0	0.0%
Disagree	0	0.0%
Moderately disagree	0	0.0%
Moderately agree	0	0.0%
Agree	3	20.0%
Strongly agree	12	80.0%

Statistics	Value
Mean	5.80
Median	6.00
Standard Deviation	0.41
% Agree broad	100.0%

The table below shows the percentage of 'Agree' and 'Strongly agree' responses to the question 'Overall I was satisfied with the quality of this person's teaching'

Overall I was satisfied with the quality of this person's teaching

Statistics	Value
% Agree	100.0%

SCHOOL: Sch Mathematics & Statistics

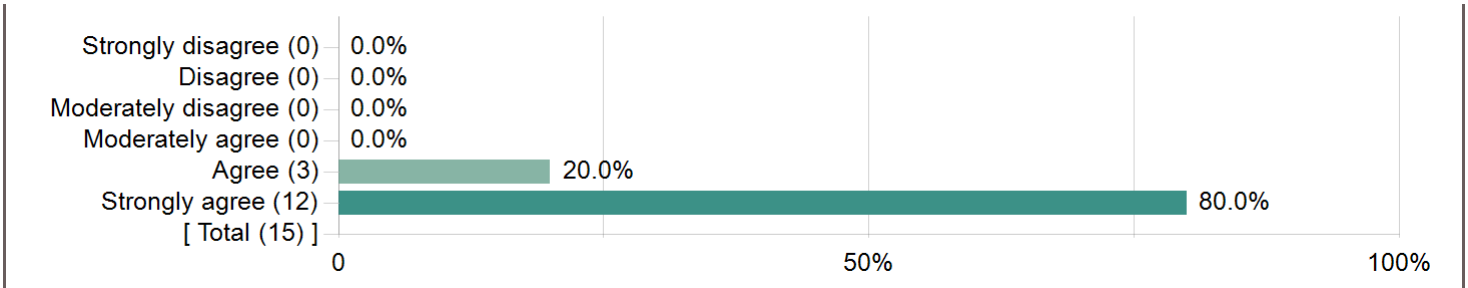
Overall I was satisfied with the quality of this person's teaching			
Options	Percentage	Statistics	Value
Strongly disagree	0.9%	Mean	5.29
Disagree	1.1%	Median	6.00
Moderately disagree	2.4%	Standard Deviation	0.93
Moderately agree	9.8%	% Agree broad	95.6%
Agree	35.2%		
Strongly agree	50.5%		

FACULTY: Faculty of Science

Overall I was satisfied with the quality of this person's teaching			
Options	Percentage	Statistics	Value
Strongly disagree	1.0%	Mean	5.30
Disagree	1.1%	Median	6.00
Moderately disagree	2.3%	Standard Deviation	0.93
Moderately agree	9.5%	Standard Error (based on SD)	0.01
Agree	34.8%	% Agree broad	95.6%
Strongly agree	51.3%		

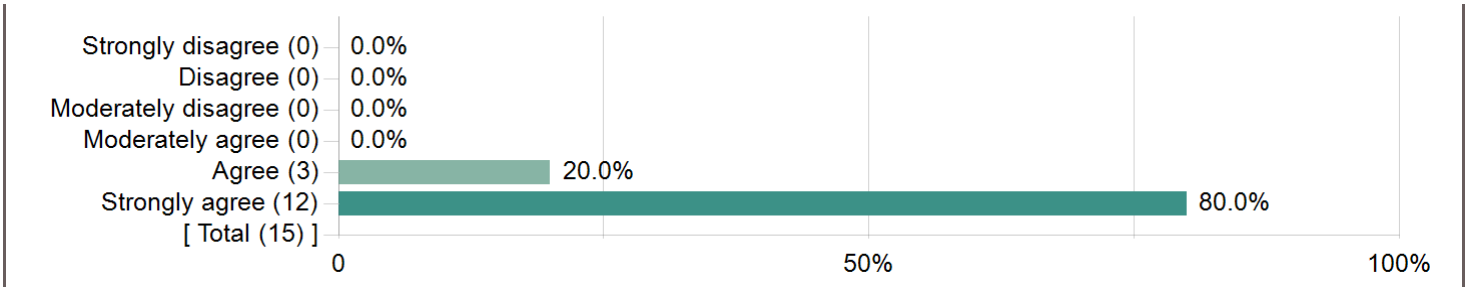
Overall I was satisfied with the quality of Balazs Elek's teaching

Overall I was satisfied with the quality of this person's teaching



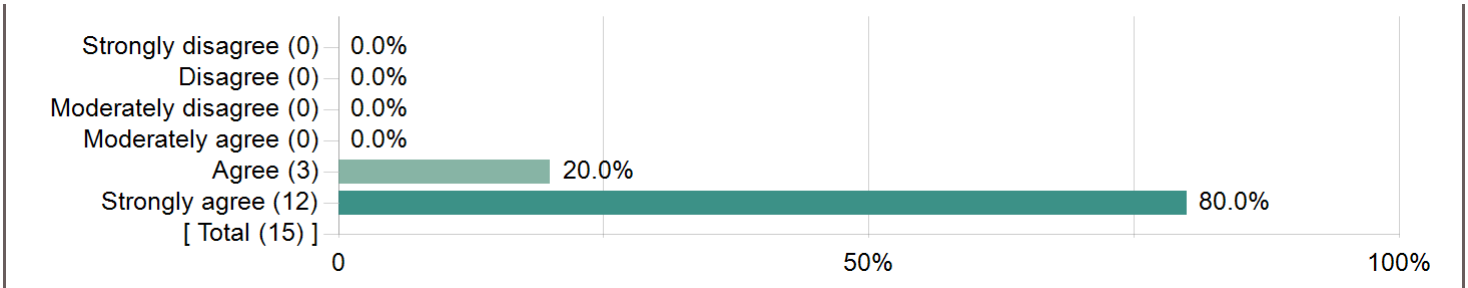
Statistics	Value
Mean	5.80
% Agree broad	100.0%

1. Balazs Elek encouraged student participation



Statistics	Value
Mean	5.80
% Agree broad	100.0%

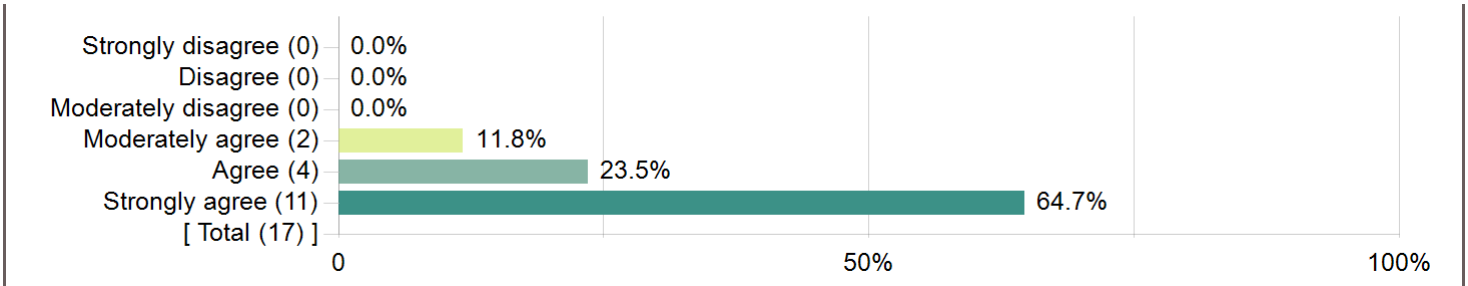
2. Balazs Elek provided helpful feedback



Statistics	Value
Mean	5.80
% Agree broad	100.0%

The weekly Activity Tutorials helped in my understanding of the course content.

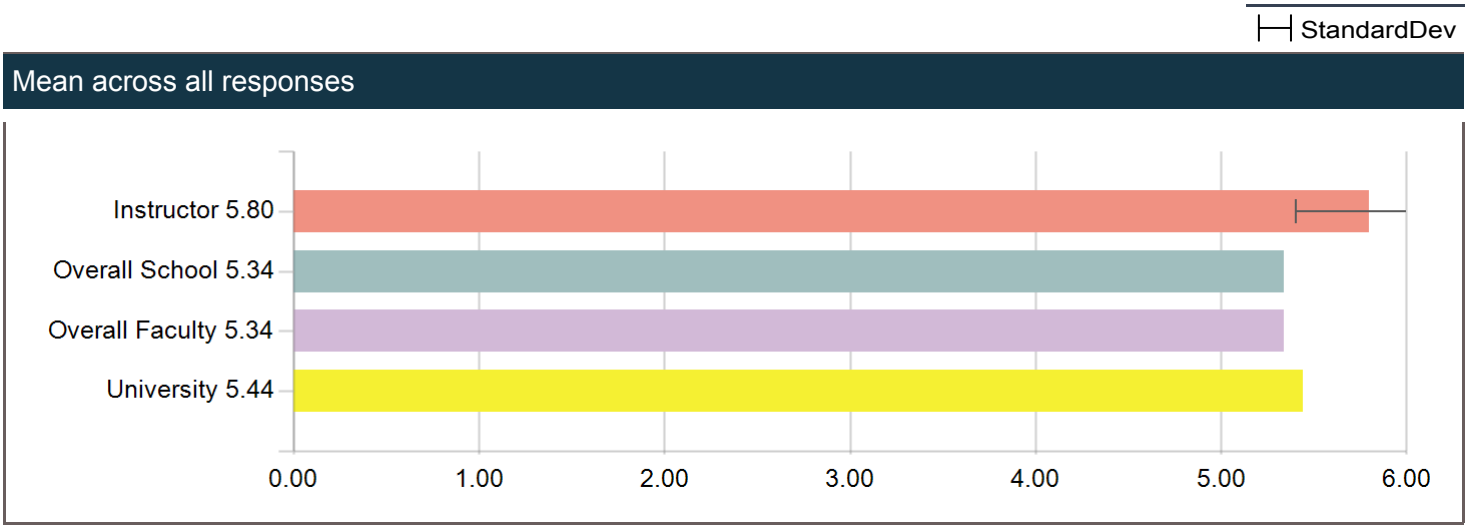
The weekly Activity Tutorials helped in my understanding of the course content.



Statistics	Value
Mean	5.53
% Agree broad	100.0%

Comparison Statistics

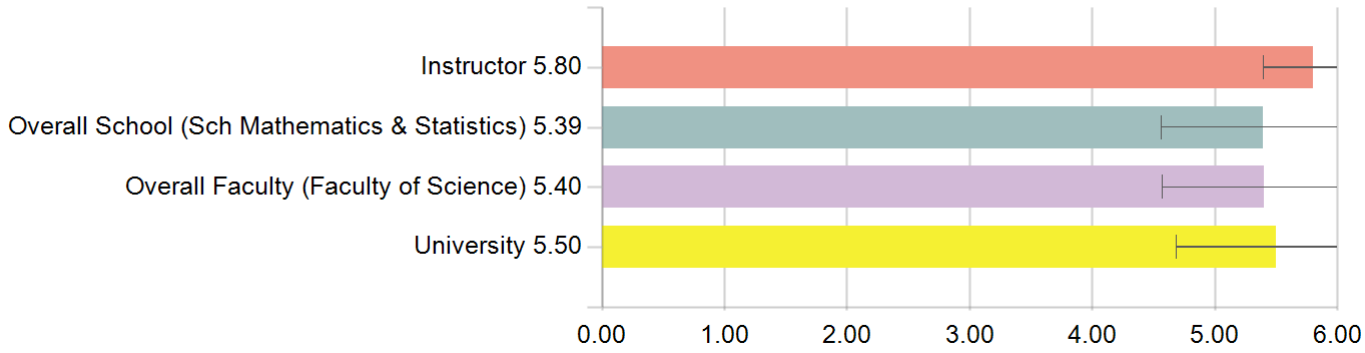
Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

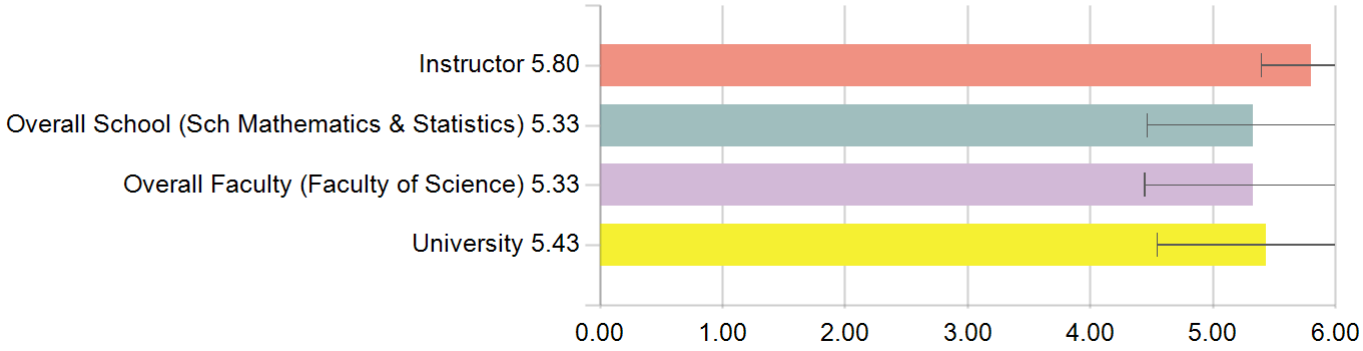
1. encouraged student participation



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

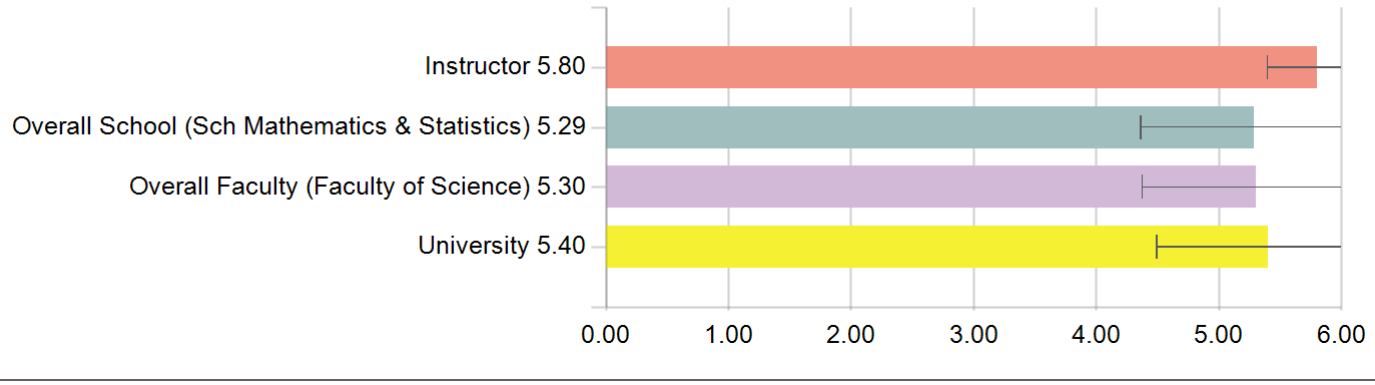
2. provided helpful feedback



Comparison Statistics (continued)

Mean (average student responses between 1 and 6) and StandardDev (Standard deviation of student responses) are used for comparison statistics between Teacher, School, Faculty and University.

3. Overall I was satisfied with the quality of this person's teaching



Raw comment data

The best features of Balazs Elek's teaching were

Comments
He carefully explained everything to us in the tutorial.
going around and making sure all students understand answers and takes all the time needed to answer questions
The professor demonstrates strong dedication and responsibility in his teaching. His lectures are well–organized, and he explains complex ideas in a clear and engaging way. He is always patient with students' questions, and his commitment helps create a positive and supportive learning environment.
He can patiently explain every problem students don't understand.
I loved Balazs's teaching!! He was so friendly and helpful, guiding us through the questions but doing so in a way which allowed us to find the answers ourselves and feel accomplished in doing so. He was a clear communicator and the maths felt much more simple when he explained it. The board tutorials with Balazs were so fun and made me feel much more prepared for the course!

THE UNIVERSITY OF BRITISH COLUMBIA

2024-25 Winter Term 1 UBC-V Individual Instructor Report for MATH_V 223-101 - Honours Linear Algebra (Balazs Elek)

Project Title: **2024-25 Winter Term 1 UBC-V Instructor SEI Surveys**

Course Audience: **73**
Responses Received: **29**
Response Ratio: **40%**

Report Comments

Recommended Minimum Response Rates

Class Size	Recommended Minimum Response Rates based on 80% confidence & ± 10% margin
< 10	75%
11 - 19	65%
20 - 34	55%
35 - 49	40%
50 - 74	35%
75 - 99	25%
100 - 149	20%
150 - 299	15%
300 - 499	10%
> 500	5%

Legend

N: Invited
n: Responded

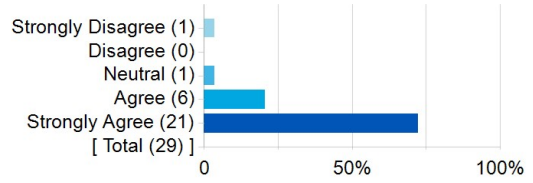
Frequency Distribution
SD: Strongly Disagree

UBC Student Experience of Instruction

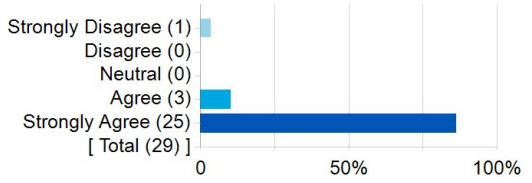
University Module Questions

University Module Questions

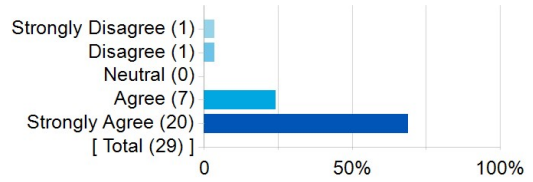
1. Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.



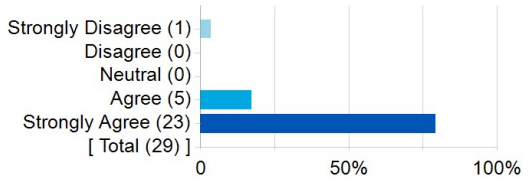
2. The instructor conducted this course in such a way that I was motivated to learn.



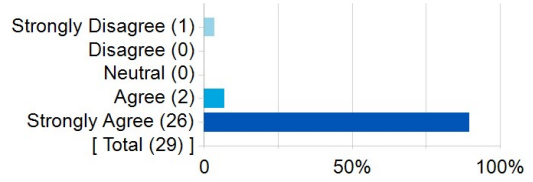
3. The instructor presented the course material in a way that I could understand.



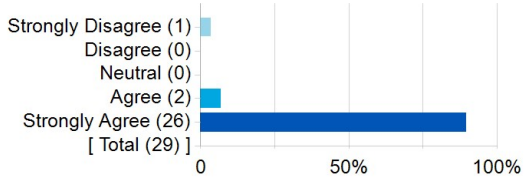
4. Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.



5. The instructor showed genuine interest in supporting my learning throughout this course.



6. Overall, I learned a great deal from this instructor.



UBC Student Experience of Instruction

University Module Questions

Question	IM	PF	DI
Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.	4.8	93%	0.3
The instructor conducted this course in such a way that I was motivated to learn.	4.9	97%	0.2
The instructor presented the course material in a way that I could understand.	4.8	93%	0.4
Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.	4.9	97%	0.3
The instructor showed genuine interest in supporting my learning throughout this course.	4.9	97%	0.2
Overall, I learned a great deal from this instructor.	4.9	97%	0.2

Faculty Questions

Course Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
My academic background provided sufficient preparation for this course.	73	29	3	3	4	7	12	0	4.1	0.7
In this class, I applied facts, theories, or methods to new problems or situations.	73	29	1	0	0	3	24	1	4.9	0.2

Question	%Favourable
My academic background provided sufficient preparation for this course.	66%
In this class, I applied facts, theories, or methods to new problems or situations.	96%

Instructor Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
The instructor treated students with respect.	73	29	1	0	0	0	28	0	5.0	0.1
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	73	29	1	0	1	5	22	0	4.8	0.3
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	73	29	1	0	0	3	25	0	4.9	0.2

Question	%Favourable
The instructor treated students with respect.	97%
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	93%
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	97%

Open ended feedback

Please comment on what your instructor did well to support your learning.

Comments
Balazs is the absolute goat. He is always available for questions and has loads of office hours. It's clear that he enjoys teaching this class and he made it really fun to work through (even though the content is quite challenging for most people).
Balazs holds 3 x 1 hour office hours each week (sometimes he extends them) to provide his students opportunities to ask/clarify everything about class content and homework. Somehow, he is always able to give hints for the homework that can tailor our thought process wihtout directly telling the answers.
Held numerous office hours and was very deliberate in making sure we had enough time to work on homework sets, providing numerous crucial extensions.
Very genuine and understanding, showed lots of care and put lots of time into helping. Also put lots of time and effort into the piazza which I really appreciated.
He was very approachable, and gave good tips for improving performance. He was also just an excellent lecturer in general—made classes very engaging. I don't think I willingly missed any of his classes, even when I was very busy with other classwork.
The instructor hosted three hours of office hours every week, often staying much longer in order for students to come together and work on problems with his assistance if needed, as well as providing a space for students to ask for direct help or clarification on any course content. Moreover the instructor was flxible with deadlines as needed following the class' advice, such as pushing back homework deadlines or no homework over the break, which really alleviated our work loads on some days.
Balazs always prioritized our understanding in lectures. If the class looked confused, were asking many questions, or suspiciously few, he would immediately explain in more depth and encourage us to state our confusions. He was very willing to extend deadlines to support students in completing the homeworks, and was very friendly and helpful in office hours. Overall, Balazs was incredibly approachable, intelligent, and witty – making lectures a genuine joy to attend.
very nice
Very enthusiastic in teaching the material.
Prof Balaz is a really great teacher. He loves teaching, and loves linear algebra
Balazs is an amazing instructor, and the way he presented the material in class was both interesting and motivated me to learn. He also held three office hours a week, which was invaluable for clarifying concepts and getting help on the homework. His class was easily my favourite this term.
Balazs is amazing. He did everything well.
Balazs Elek explains every thing really well
Balazs did a great job both at conveying the (very difficult) course content, and doing so in a way that was fun and engaging.
Incredible lecturer, great at rephrasing and answering questions. Great content and well–structured homework which require good understanding, with a great support network between office hours and piazza. Sad to see lectures end, hope to take more courses with dr Balazs in the future.
Balazs is probably the most inspiring instructor I have ever encountered. I am fairly new to the field of mathematics, especially at this level, but the way the course was run, made the transition seamless. I recall doing fairly poor on one of the midterms, even though I studied relentlessly in preparation, to say the least I was considering dropping the course. I asked the professor for advice on how to go about the course and the answer I received was something along the lines of, focusing more on learning then being fixated on grades and how test environments are a poor measure for my mathematical ability. This doesn't seem like quite a response, but to me it was literally what kept me in the course. I decided to step away from habits that would conventionally help me do well in other classes and focus on "the bigger picture". Balazs would hold multiple office hours throughout the week and that in of itself made me sure of becoming a math major. Balazs would encourage collaboration among his students, dropping subtle hints and having the students discuss among themselves about how to solve a particular problem. I would say the classroom culture of this class is unlike anything I have really experienced, I am quite introverted and was not keen to attend office hours (still am not), but particularly for this class I make the exception. Most instructors focus purely on learning from the textbook and having their students mindlessly regurgitate what they had read, but in this class we are asked to question and argue everything we read. In our piazza/classroom we have students not accepting what is taught to them and I feel like that is one of the biggest indicators that students are actively learning. This is probably the hardest class I have taken to date, but it is also the most I have ever been engaged. Balazs does not limit the scope of our learning to what is in the course outline, but whatever the student wishes to learn

UBC Student Experience of Instruction

Comments
and I think that's amazing.
Professor Balazs Elek has been an incredible instructor this term across the board. Despite the difficult course material Professor Elek always clearly communicated the material in engaging lectures with helpful examples. Additonally, throughout each lecture Professor Elek consistently makes sure to check in with the class regarding how well we are following the lesson and if we understand what he is showing us. Aside from the lectures Balazs provides us with weekly homeworks. Although they are very difficult at times, I have found them to be quite interesting and I truly believe that they have been very helpful with my understanding of the material. Outside of class he is also very accessible with flexible and consistent office hours as well as frequent responses on Piazza. In general, it is clear that Professor Elek has been doing his absolute best to create a welcoming and effective learning environment that has allowed myself and my classmates to feel comfortable with asking questions and seeking clarification. Overall, the combination of Professor Elek's effective teaching style and kind attitude have easily made Math 223 my favourite class this term and given me a deep appreciation for the material. I really can not overstate how excellent of an instuctor Professor Elek has been as he is easily one of the best teachers I have ever had.
best math prof i have EVER had at UBC (objectively — i can prove it if you'd like). Thanks for an amazing semester!!!!
The professor had a lot of office hours, he was very active on piazza and would usually stay behind after class for a very long time until I understood what I was asking. He was also very enthusiastic with the course, pacing back and forth across the board and having fun little story time about math which made me always pay attention.
I really enjoyed his humor and energy in class which made me more engaged to learn in class. The weekly homeworks were very hard and challenging but ultimately helped me take time to understand the abstract ideas.
Relate new concepts to previous concepts, provide motivations for the stuff we're learning.
The instructor is very patient and supportive when answer our questions. I really appreciate it!
He made sure to explain everything well and showed his work when explaining concepts.
He cares for the students well, patiently answers all the class questions, and has many office hours every week.
The instructor was very caring and slowed down lectures when he noticed that the class was having trouble understanding a concept. The instructor also emphasized concepts that were important by explicitly stating that they were, or going through proofs slower so that everything is as understandable as possible.

Do you have any suggestions for what the instructor could have done differently to further support your learning?

Comments
N/A
Maybe I just expect him to go at a constant pace throughout the term because we used to go in a slow pace in week 3–5 but ended up rushing at the last week, but I think this is still very tolerable since this is the first time Balazs teaches MATH 223 and the content itself is very advanced.
Lectures were sometimes a bit difficult to follow. Some extra examples of applications of the theorems would be helpful.
Nope!
The homework assignments were very time consuming for me, and detracted from the time I had available to study for this and other courses. Balazs was very accommodating with due dates, which was appreciated, but I would have preferred a somewhat shorter homework.
I think the only thing that could be improved is having access to instructor's class notes. that being said the content did stick to a set textbook so it was also easy to follow from there, as the instructor did note what sections were covered, but being able to fill in any gaps betwen the textook and the course could be more useful, but not necessary.
Some form of posted class notes would be appreciated, as we did have to rely on hand taking notes from the chalkboard.
no
I think it would be helpful to be more strict on assignment deadlines, as I fould it hard to get into a routine when the deadlines changed week–to–week
While I did appreciate his flexibility in extending the homework deadline when we needed it, it did create quite a bit of uncertainty on how many homeworks we would have and how we would be evaluated at the end of the term. I would have preferred a bit more consistency in that regard.
Perhaps more emphasis on why we learn things before delving into details and proofs would be beneficial.
N/A
No, I think he did a great job.
Keep on building strong connections with your students and bringing personality to every lecture, it is extremely engaging.
I believe Professor Elek taught his class as well as he could all things considered.
i think it wouldnt have hurt to be a little stricter on the hw deadlines. honours students can work under pressure! we get lazy when we dont have to lol
none
I think it would benefit to first start with writing down all the theorems we would go over in class and then proving instead of going through each theorem and proof one–by–one. I understand that sometimes you need a theorem to prove another theorem, but knowing all the ideas we'd go over before we start proving them could help us grasp what we are going over in class. I also would have enjoyed more work on the computational side of linear algebra. I also believe that one week is not enough time to finish the homework given due to how challenging it is (I'm a combined honours major so I also had a lot of work) so I think maybe a bit more time to work on homeoworks would alive stress and also allow future students to properly work on each question.
I wish there could've been more office hours. I'm very appreciative of how responsive he was on Piazza, but in–person learning would be more effective for me. I would've went to them every time if it fit my schedule.
I do not.

UBC Student Experience of Instruction

Please identify what you consider to be the strengths of this course.

Comments
Super useful for pretty much anyone pursuing a career in a science–based career.
This course teaches students to develop their abstract thinking which will be helpful for understanding any other disciplines, such as physics.
Proof–based approach to linear algebra allows for deep understanding of the subject
Interesting, generally good pace, good homeworks, well taught
The professor (Balazs)
As mentioned above, having a set textbook to refer to whenever something was not clear from class, in addition to a list of chapters covered when was really useful when information wasn't clear from class. Moreover, although this class had a very heavy work load, working through the weekly homework did help in understanding and clarifying the sometimes extremely abstract and hard to comprehend material, as well as allowing us to find reasons or applications to the materials learned in class (in opposite to just being taught some abstract mathematical theorem and having no idea why that would be useful ever)
Having few interesting and challenging homeworks support learning and keep the course more engaging, as opposed to many webworks.
fun
Teaching linear algebra in a more in depth way.
I think it's an interesting and very useful introduction to linear algebra. I really enjoyed the course and its content overall.
Everything.
In general, I found the lectures to be the best part of the course.
The semi–weekly long homeworks forced you to understand the material in the moment, making procrastination difficult which was incredibly useful. Though they could be tedious, there was a large support network and a professor with excellent office hours and an aptitude for understandably answering questions.
Instructor Textbook (Axler) Frequency of office hours Homeworks in this class are made in a very interesting way, the questions are extremely irritating at times, but equally fun.
I think having a somewhat small class size allows lectures to be more engaging and was overall a good thing this term.
engaging material, interesting lectures, fun homeworks
Understanding the ins and outs of linear algebra in an abstract way
I think this course is really beneficial for critical thinking and abstract thought around math.
Mathematical thinking and understanding fundamental properties of widely used math.
Everything about this course was wonderful but particularly the content of the course was extremely interesting.

UBC Student Experience of Instruction

Please provide suggestions on how this course might be improved.

Comments
Maybe spend a bit more time on computation at the beginning to have a more sound basis for our understanding, as most people have done no linear algebra before this course.
This course is already well–designed.
Having more opportunities to apply the concepts to solving computational problems with linear algebra would be quite useful to see how the concepts are applied in the real world. Even if it is just a set of optional practice problems.
Pace was a little quick at the end but that's it
Introduction to proofs early on in the course, since this is the first proof–based course for many students. Personally, I didn't understand what a proof was for a decent chunk of time at the start of the course, which was not great because we had assignments very early on. For example, some things that I wish had been covered in the first class or so: – What is a proof and why do we need one? – What does if and only if mean? Why do we need to prove it in two directions? – What is a proof by induction and how do you do one? (step by step) A lot of these things we were expected to learn by doing, which may have suited some students, but as someone who has not done proof–based math before in any capacity, I struggled very much with understanding.
no suggestions
The midterm grading distributation is fair, though for the difficulty the homeworks should be worth more.
no
I understand that using latex is an important skill, but requiring that every homework be typed up was a lot of work, especially at the end of the term when the homework had 9 questions instead of the usual 4–5 and we all had finals to study for.
Nothing.
Too many homeworks :–(Maybe make one due every two weeks instead of sporadically every week. The homeworks are too hard to do in one week
Make the last homework a bit easier.
Again, I don't have any suggestions
Questions that are non–graded, (as the homework is quite difficult alone) such that they are more for the student to ponder over. Linear algebra is massive topic and I feel like it would be nice to get questions that students could ponder over, that possibly we will not be able to solve even by the end of the term. I feel like this would dive students into the idea that solving problems in higher–mathematics in reality isn't fixated into a week.
I think having the class in the Forestry Building was quite an inconvenience for the majority of the class as the majority are in some sort of math or physics program and the major physics and math buildings are quite far to commute in 10 minutes. Additionally I think more TA support could have been helpful as well.
It would be nice if the course can go a little slower
I wish there was more content. To be fair to the instructor, the course did go a lot faster over material than almost every other course I have had, but I think there could have been a little more content added by shortening some sections. I also recognize that there is a limit to the amount of content this course can cram in without being excessively quick.

THE UNIVERSITY OF BRITISH COLUMBIA

2024S UBC Individual Instructor Report for MATH 100 006 - Differential Calculus with Applications (Balazs Elek)

Project Title: **2024S UBC Instructor SEI Surveys**

Course Audience: **23**
Responses Received: **4**
Response Ratio: **17%**

Report Comments

Recommended Minimum Response Rates

Class Size	Recommended Minimum Response Rates based on 80% confidence & ± 10% margin
< 10	75%
11 - 19	65%
20 - 34	55%
35 - 49	40%
50 - 74	35%
75 - 99	25%
100 - 149	20%
150 - 299	15%
300 - 499	10%
> 500	5%

Legend

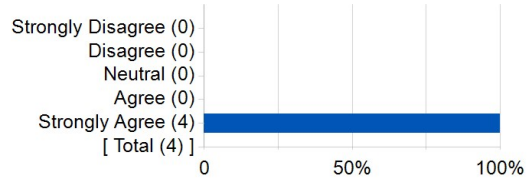
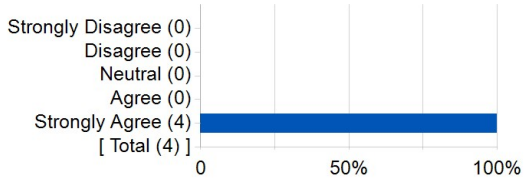
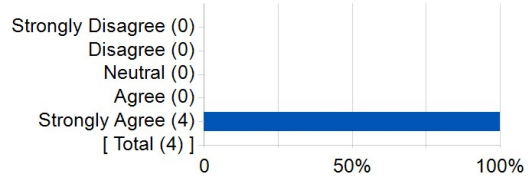
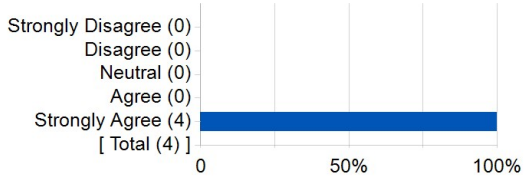
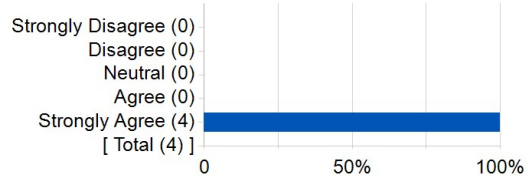
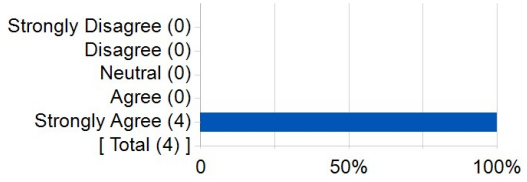
N: Invited
n: Responded

Frequency Distribution
SD: Strongly Disagree

UBC Student Experience of Instruction

University Module Questions

University Module Questions

1. Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.	2. The instructor conducted this course in such a way that I was motivated to learn.
	
3. The instructor presented the course material in a way that I could understand.	4. Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.
	
5. The instructor showed genuine interest in supporting my learning throughout this course.	6. Overall, I learned a great deal from this instructor.
	

UBC Student Experience of Instruction

University Module Questions

Question	IM	PF	DI
Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.	5.0	100%	0.0
The instructor conducted this course in such a way that I was motivated to learn.	5.0	100%	0.0
The instructor presented the course material in a way that I could understand.	5.0	100%	0.0
Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.	5.0	100%	0.0
The instructor showed genuine interest in supporting my learning throughout this course.	5.0	100%	0.0
Overall, I learned a great deal from this instructor.	5.0	100%	0.0

Faculty Questions

Course Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
My academic background provided sufficient preparation for this course.	23	4	0	1	0	2	1	0	4.0	0.6
In this class, I applied facts, theories, or methods to new problems or situations.	23	4	0	0	0	1	3	0	4.8	0.2

Question	%Favourable
My academic background provided sufficient preparation for this course.	75%
In this class, I applied facts, theories, or methods to new problems or situations.	100%

Instructor Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
The instructor treated students with respect.	23	4	0	0	0	0	4	0	5.0	0.0
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	23	4	0	0	0	0	4	0	5.0	0.0
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	23	4	0	0	0	0	4	0	5.0	0.0

Question	%Favourable
The instructor treated students with respect.	100%
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	100%
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	100%

Open ended feedback

Please comment on what your instructor did well to support your learning.

Comments
A small detail, but he used coloured chalk when teaching, which not only helped in everybody's understanding but from taking this course previously, I know he brought in especially which showed to me his care in teaching. When asking a general question on something I may be stuck on, he always was able to find through his questions the area I was struggling with specifically which really helped me target concepts I should improve on. He is kind and very welcoming, and makes class fun and interactive which can be rare in university. Overall he is one of the best teachers I have had and he is one of the sole reasons I have the courage and motivation to keep pursuing science.
One of the best professors I have ever had, he goes above and beyond to help students succeed. His dedication to teaching and genuine passion for the subject are evident in every small class. He consistently stays extendedly after class hours, taking the time to explain concepts in depth and breaking them down into understandable parts. This approach not only helps in grasping difficult topics but also increases my love for math. His commitment to ensuring that each student fully understands the material makes learning enjoyable and inspiring. I would love to take every math class with him, as his teaching style and support have significantly enhanced my academic experience. Balazs is truly a gem!!
Balazs would consistently stay after class to answer questions whether they be conceptual or even textbook questions. The way he explained things always seemed to click with me.

Do you have any suggestions for what the instructor could have done differently to further support your learning?

Comments
It would be great to have access to the small class notes we made that we give in after class, or a revised conclusion sheet on Canvas.
You teach with so much passion! Don’t lose that.

Please identify what you consider to be the strengths of this course.

Comments
Expectations are clear, office hours are plenty, instructors really cared.
This course is very comprehensive on the uses of calculus and its principles. I felt like I understood its real–world impacts and uses as well as how some of these discoveries were made. The practice midterms and homework are very similar to the midterm and finals which help feel that the work you put in you truly get out. Lots of resources available, a good amount of office hours and an organised flow of learning.

Please provide suggestions on how this course might be improved.

Comments
I think assignments should be weighted heavier given how much effort goes into them. Possibly weight the final less as well.
The written assignments are my main concern. Almost every group had 1–2 missing members but it felt unaddressed. The "un–contributing" label wasn't the most helpful as adding it feels like starting unnecessary drama and that your peers will hate you and may not help in future. Also due to this, it took too much time to complete these assignments. I would also prefer if they were slightly less applied in their concepts as although it adds breadth to my learning, this course is difficult enough that I would prefer more assignments that acted like 3 or 4–star exam questions, not one long situational problem that may use similar techniques.

UBC Student Experience of Instruction

Explanatory Note

The reported metrics are as follows:

1. Percent Favourable Rating

This is the percentage of respondents who responded with a 4 or 5 (Agree or Strongly Agree) on a scale of 1 to 5.

2. Interpolated Median

The data collected for Student Experience of Instruction (SEI) are ordinal in nature, with a natural order (from 1 to 5). The usual measure of central tendency for ordinal data is the median (50% percentile). The Interpolated Median (IM) is an adjusted median that considers the number of responses less than the median, greater than the median and equal to the median. As such, IM reflects the distribution of students' responses.

Consider the following example:

Frequency Distribution

Response for University Module Item	Section 1	Section 2
5 = Strongly agree	5	5
4 = Agree	3	5
3 = Neither agree nor disagree	6	0
2 = Disagree	1	2
1 = Strongly disagree	0	1
Mean	3.8	3.8
Median	4.0	4.0
Interpolated Median	3.7	4.2
Percent favourable rating	53%	77%

THE UNIVERSITY OF BRITISH COLUMBIA

2023W2 UBC Individual Instructor Report for MATH 101 B15 - Integral Calculus with Applications (Balazs Elek)

Project Title: **2023W2 UBC Instructor SEI Surveys**

Course Audience: **58**
Responses Received: **11**
Response Ratio: **19%**

Report Comments

Recommended Minimum Response Rates

Class Size	Recommended Minimum Response Rates based on 80% confidence & ± 10% margin
< 10	75%
11 - 19	65%
20 - 34	55%
35 - 49	40%
50 - 74	35%
75 - 99	25%
100 - 149	20%
150 - 299	15%
300 - 499	10%
> 500	5%

Legend

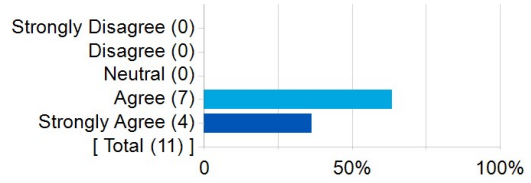
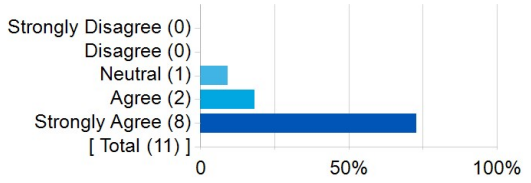
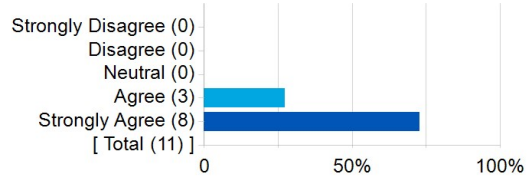
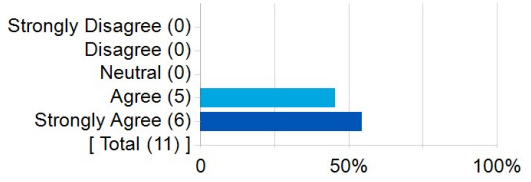
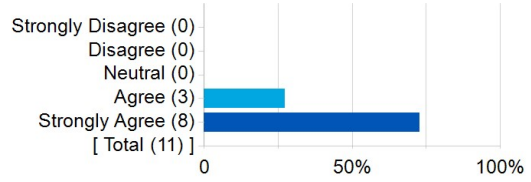
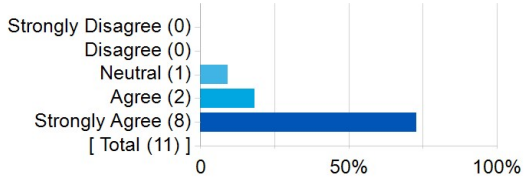
N: Invited
n: Responded

Frequency Distribution
SD: Strongly Disagree

UBC Student Experience of Instruction

University Module Questions

University Module Questions

1. Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.	2. The instructor conducted this course in such a way that I was motivated to learn.
	
3. The instructor presented the course material in a way that I could understand.	4. Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.
	
5. The instructor showed genuine interest in supporting my learning throughout this course.	6. Overall, I learned a great deal from this instructor.
	

UBC Student Experience of Instruction

University Module Questions

Question	IM	PF	DI
Throughout the term, the instructor explained course requirements so it was clear to me what I was expected to learn.	4.3	100%	0.2
The instructor conducted this course in such a way that I was motivated to learn.	4.8	91%	0.3
The instructor presented the course material in a way that I could understand.	4.8	100%	0.2
Considering the type of class (e.g., large lecture, seminar, studio), the instructor provided useful feedback that helped me understand how my learning progressed during this course.	4.6	100%	0.2
The instructor showed genuine interest in supporting my learning throughout this course.	4.8	100%	0.2
Overall, I learned a great deal from this instructor.	4.8	91%	0.3

Faculty Questions

Course Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
My academic background provided sufficient preparation for this course.	58	11	0	2	3	2	4	0	3.8	0.6
In this class, I applied facts, theories, or methods to new problems or situations.	58	11	0	1	1	4	5	0	4.4	0.5

Question	%Favourable
My academic background provided sufficient preparation for this course.	55%
In this class, I applied facts, theories, or methods to new problems or situations.	82%

Instructor Questions

Question	N	n	SD	D	N	A	SA	N/A	IM	DI
The instructor treated students with respect.	58	11	0	0	0	2	9	0	4.9	0.1
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	58	11	0	0	0	4	7	0	4.7	0.2
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	58	11	0	0	0	3	8	0	4.8	0.2

Question	%Favourable
The instructor treated students with respect.	100%
The ways the instructor implemented the course activities (e.g., in-class activities, labs, tutorials, field trips, online components, assignments) helped me achieve the learning objectives.	100%
The instructor was intentional about cultivating a welcoming and inclusive environment that supports all students and encourages all students to participate.	100%

Open ended feedback

Please comment on what your instructor did well to support your learning.

Comments
The instructor was very patient in office hours and with the group, and allowed one to present their thoughts about how to approach a question without feeling pressure to get it right. They were always very engaged in lectures which helped to motivate me to learn and understand the material. They made themselves very approachable at office hours, and I really appreciate them taking the time to answer all of our questions completely in a way that is easily understood.
Very nice instructor.
Very in depth explanations and constantly helped small groups with challenging material with walkthroughs and explaining new methods of thought
He is the best. He made all of the course material very easy and fun to learn.
Balazs is a wonderful instructor, he always makes everything make sense and will check for understanding.
He's great at explaining concepts by breaking it done into smaller pieces, awsome TA!
Always helped our group to get back on track when we were off. His explanations were very clear and easy to understand.
Very engaging in class. Provided feedback to students when we were working on in–class worksheets. Went through the worksheets as a class and tried to explain concepts so that everyone would understand.
Engaging and tries to teach in a concise and efficient manner as much as possible even though they are forced to teach fast.

Do you have any suggestions for what the instructor could have done differently to further support your learning?

Comments
In the small classes, it would have been nice to go over some harder examples to better prepare us for the webwork and written assignments.
Consider to cancel the written assignment.
No
None
There were some classes that we did not get through all the worksheet material.
No

UBC Student Experience of Instruction

Please identify what you consider to be the strengths of this course.

Comments
The small classes provide the opportunity to actively test your understanding of a concept, which helps to identify which areas are your strengths and weaknesses. The instructor has the opportunity to discuss the problems and fundamentals with students directly, as well as facilitate group discussions. The increase in instructor–student communication in the small class portion is significant in helping students develop an understanding of how to apply their mathematical knowledge to different problems.
Hard.
The small classes
Webwork
It's important for future studies involving math.
The small class and getting hands on experience with the course content.
I like the webworks
No strengths at all in small classes. Useless to teach a new math concept in one hour and expect everyone to understand anything. In one small class session, even the instructor mentioned how "we're supposed to remember Taylor Series, but let's be real, no one remembers it." This course puts too much unrealistic expectations and fails to meet reality.

Please provide suggestions on how this course might be improved.

Comments
It would be nice if the written assignments were not always due the same week as the test, as this made it difficult to properly complete it. Furthermore, if your group did not contribute, the written assignments were not manageable for one person, which led to stress, poor grades, and overwhelm for the solitary contributing student. Either written assignments need to be spaced out more, be worth less of the final grade, or students need to have some control over their groups particularly if the other members are not contributing.
Consider to cancel the written assignment.
The tests are too hard. They strive to make people fail. My classes average was well below passing on both tests, that should be a surefire sign things are not being taught but being put on the test. We never once go through anything above a 2 star question in class making it nearly impossible to understand a thing. The writing on the written assignments is completely ambiguous leaving most students asking each other how to figure it out or turning to the internet
none
two small class times a week would be ideal.
Small class answers are not posted.

Cornell University
Course Evaluation Response Summary
Semester: Spring 2021 Course Owner: MATH
Course: MATH 1910 LEC 001 CID: 4453
Title: Calculus For Engineers
Instructor: Elek
20 Responses, 37 Enrolled, 54.05% Response

13. If you have any comments about the instructor, please include them here.

13234. Best math instructor I have had thus far.

12180. Balazs Elek is the best instructor I've had at Cornell so far, by far as well. He does a great job of engaging with the class during lecture, even with the zoom / COVID situation, by asking questions and confirming that nobody is overly confused with the material at hand. His office hours are extremely helpful as well; I went to his office hours almost every time they were held. From homework help to practice exam questions, Balazs' office hours provide the best support. Three adjectives to describe Balazs: enthusiastic, helpful, empathetic. After my fall semester math class being such a disaster and my high school math classes being lackluster, Balazs made me love and appreciate math again.

13147. Very helpful. I dont really think I do well but he still inspires me to try to understand and not give up. Breaks down difficult concepts to aid my understanding

18255. Sometimes the instructor got lost on the explanations which made it a little bit harder to understand the subject.

30045. Professor Balazs was wonderful, caring, and an overall very helpful and considerate instructor. He taught the material that would be tested on the exams and was very helpful during office hours. I will surely miss his instruction.

18207. The professor had a different style of teaching than most of the instructors that I have seen at this college so far. He is really invested in ensuring that his students can understand the material. He is willing to meet with you outside of class regardless of how simple your question and regardless of your performance of the course. However, more importantly, he really and truly cares about the well-being of his students and does not see us as machines who began his class with the same level of knowledge. Professor Balazs understands that his students are human and that the things that are happening around them and at the colleges and dorms/homes can truly affect them. Not only is he willing to listen to his students, but he is more than willing to be accommodating, and I am so very thankful for him. I am thankful for his considerate heart and his continuous support and guide with various study techniques and tips.

5208. He was very approachable and considerate of our difficult circumstances this semester. I have never felt overwhelmed by this class though it is still challenging as math classes should be.

Cornell University
Course Evaluation Response Summary
Semester: Spring 2021 Course Owner: MATH
Course: MATH 1910 LEC 001 CID: 4453
Title: Calculus For Engineers
Instructor: Elek
20 Responses, 37 Enrolled, 54.05% Response

14. Please comment on any other aspect of this course, including the lecture, textbook(s), homework, exams, course content, etc. Would you recommend this course to a friend?

(No comments about the instructor in this box, please.)

13234. I would recommend this course to a friend.

12180. I would definitely recommend this course to a friend. The exams are a little bit harder than they should be, but not to the point where they're super unfair or anything.

18255. The material that the professor prepared for the class were very high quality. They were very well prepared and definitely allowed me to better comprehend each subject.

5208. I would definitely recommend this course to a friend if they needed it. Based on this class I would suggest that people take the engineering math courses over other math classes. I took MATH 1110 last semester and did not feel like I could get the help or attention I needed. The staff seemed to be more concerned with not making our grades too high than our actual learning experience.

FAS Summer 2020 Undergrad F

Course Name: Linear Algebra I MAT223H1-F-LEC0201
Division: ARTSC
Session: F
Session Codes: F = First/Fall, S = Second/Winter

Instructor: Balazs Elek
Section: LEC0201

Report Generation Date: July 8, 2020

Raters	Students
Responded	32
Invited	125

Section 1: Course Evaluation Overview

Part A. Core Institutional Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
I found the course intellectually stimulating.	4.1	4.0
The course provided me with a deeper understanding of the subject matter.	4.2	4.0
The instructor (Balazs Elek) created an atmosphere that was conducive to my learning.	4.4	5.0
Course projects, assignments, tests, and/or exams improved my understanding of the course material.	4.1	4.0
Course projects, assignments, tests and/or exams provided opportunity for me to demonstrate an understanding of the course material.	3.7	4.0
Institutional Composite Mean	4.1	-

Scale: 1 - Poor 2 - Fair 3 - Good 4 - Very Good 5 - Excellent

Question	Summary	
	Mean	Median
Overall, the quality of my learning experience in this course was:	3.8	4.0

7. Please comment on the overall quality of the instruction in this course.

Comments
For an online course, the quality of lectures were very good and aided in my understanding of the concepts by forcing me to apply knowledge that I learned prior to the class. However, I found many of the extra videos tedious and hard to get through, especially the ones that were 30–50 minutes in length.
I am so impressed with the way this course was delivered in an online format and actually feel that I got a better education as a result of the course being online. Professor Balaz did a great job at interacting with the students online and answering questions in a respectful and clear way. I really enjoyed the structure of lecture!
I enjoy this online course. The quality of the notes isn't as good nor organized. This is mainly down to the software we're using. I don't particularly enjoy Zoom but then again there isn't a better app at the moment.
Professor Belaz explained steps to solve questions well. He uses technical tools to show 3D space to us and uses that to help us understand the concepts better, which is really useful and helpful!
good
instruction was not great, there was not much instruction at all, all class time was used for group discussions instead of instruction from the professor, the instructions that were given in class were technical and computational that were very different from the explanation questions given on the tests and did not help to prepare us for the tests in any way, it was very unfair for the all the questions on all the tests to be explanation questions!
Very good quality.
The course is well structured and the objectives are very clear.
The overall quality of the instruction in this course was excellent and accomodating to the way we learn. For example, when we had questions, I had many opportunities to ask those questions during lectures.
Pretty good, but tests and quizzes are bit hard
While I was skeptical of the perusall assignments at the start of the course, the assignments were very helpful in actually learning the material of the course, not only because one could ask questions but in order to make thoughtful and useful comments, one really had understand the material fully and be able to think back and connect it previous problems. So the assignments encouraged the sort of deeper learning which I found helpful. I liked that while class time was all questions, all the questions helped discover links between the concepts. Overall, the quality and style of teaching was very conducive to learning. Also the course was well organized, which is always nice. And, when there was a change in schedule, they were very open about it and was not made harder on the student.
great
Prof. Elek excels at explaining concepts in a lucid manner. He is also extremely kind when he has to repeat something, or explain something again, as was often the case. He always stayed after class time to answer questions, and he never became impatient with anybody, although there were many extremely inane inquiries posed by people who had come to class without having understood the reading. He facilitated an excellent atmosphere in which to learn. I also appreciated that he had the good sense to give us a twenty–minute break in between, because a three hour class is more than most people can take.
The instruction is alright. But the tests and the exams are too difficult.
Tests/exam questions wordings are hard to understand
Well–around and accurate
The instruction was really good. The instructor and the TAs explained the material well.
Great Instructor! Assumed no base knowledge and taught everything from ground root levels!
Average
I like the instruction but I felt that we could have gone a bit faster on certain topics that i felt most people understood and only a few didn't
he accepted the advice from students to enhance the quality of lecture like give the answer of polls that was good, but the quality of his course needs to be improved, he can't explain a question that well and didn't go deep of the definition I believe the time of the lecture is a part of the reason but the way he explains a question should be more clear, instead of assuming most of us know the how to solve some kind of problems.

8. Please comment on any assistance that was available to support your learning in this course.

Comments
I did not participate in office hours but Prof Elek provided assistance by spending extra time after class to answer some of my questions which I found very helpful.
Piazza was always available to us and was very useful. As well because of the way we did textbook readings, we were able to ask questions about the readings and have other students answer. Additionally, Professor Balaz was always available for office hours (however I did not attend).
Office hours are absolutely amazing for this course. There are so many options to choose from and I benefited significantly from them. I suppose this virtual office hour works better than in person office hour given how convenience it is.
Pizza is definitely a good tool for learning. Instructors and TA generally reply to our questions very quickly.
good
office hours with professors and tas
Very helpful
Amazing TAs, breakout sections and annotations are helpful strategies.
I appreciated that there was help in the Math Help Center for the course.
There was a lot of extra worksheets and tutorial questions which was provided which was helpful. Piazza was there, and the professors responded quickly and helpfully. I knew there were office hours but I didn't go to them, but they were well advertised. The professor always stayed a bit after lectures which was helpful. Overall, while I didn't make full use of assistance in this course, I always knew that it was there and where to look.
office hour
N/A
Office hours, piazza
Office hours and talks required by students after the lectures
The office hours were really helpful.
There were plenty of office hours and quick responses on piazza

Part B. Divisional Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
FAS001 The instructor (Balazs Elek) generated enthusiasm for learning in the course.	4.6	5.0

Scale: 1 - Very Light 2 - Light 3 - Average 4 - Heavy 5 - Very Heavy

Question	Summary	
	Mean	Median
FAS002 Compared to other courses, the workload for this course was…	3.9	4.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - Strongly

Question	Summary	
	Mean	Median
FAS003 I would recommend this course to other students.	3.9	4.0

Part C: Departmental Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) explained concepts clearly.	4.3	4.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) was approachable.	4.6	5.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) answered questions clearly.	4.3	5.0

Scale: 1 - Poor 2 - Fair 3 - Good 4 - Very Good 5 - Excellent

Question	Summary	
	Mean	Median
UNIT(OQI) Overall, the quality of instruction provided by (Balazs Elek) in this course was:	4.3	5.0

FAS Fall 2019 Undergrad

Course Name: Intro Combinatorics MAT344H1-F-LEC0201
Division: ARTSC
Session: F
Session Codes: F = First/Fall, S = Second/Winter

Instructor: Balazs Elek
Section: LEC0201

Report Generation Date: January 21, 2020

Raters	Students
Responded	34
Invited	77

Section 1: Course Evaluation Overview

Part A. Core Institutional Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
I found the course intellectually stimulating.	4.1	4.0
The course provided me with a deeper understanding of the subject matter.	4.1	4.0
The instructor (Balazs Elek) created an atmosphere that was conducive to my learning.	4.2	5.0
Course projects, assignments, tests, and/or exams improved my understanding of the course material.	4.2	4.0
Course projects, assignments, tests and/or exams provided opportunity for me to demonstrate an understanding of the course material.	4.2	4.0
Institutional Composite Mean	4.1	-

Scale: 1 - Poor 2 - Fair 3 - Good 4 - Very Good 5 - Excellent

Question	Summary	
	Mean	Median
Overall, the quality of my learning experience in this course was:	4.0	4.0

7. Please comment on the overall quality of the instruction in this course.

Comments
very good
Course was pretty good in general, but it was v slow on the beginning with the basic content, and pretty fast in the end with the hard content
Good lectures
He's very helpful, appreciates (and stimulates) questions, open to office hours. Overall, excellent, highly recommended.
Prof Elek is super nice and does really take the responsibility seriously. Super kindful
The instruction is clear and interesting, but it was a little confusing for students had no experience in problems involves probabilities in the beginning.
Intuitive lectures, well explained
It's just a well run, organized course on cool material. I have zero complaints. This is not an empty compliment, I usually have many complaints. But good work on this one. It was fun.
Henry and Balaz are both very good instructors. notes are very clear and good for review. Course content is not only interesting but also requires many practice. So really a good course.
good
It was moderate in all senses. Although the class is really hard I think the instructors and TAs helped us good
Great instructor, explained concepts in depth and elaborated/clarified on any points that were unclear.
The class went pretty well and gain a lot of knowledge from it.
My professor was very nice and he prepared lectures very well. He was also very generous and kind to give us help and support. I enjoyed this course a lot.
Material was presented very clearly and room was left for thinking and attempting solutions. Course notes were very high quality, very easy to read, and contained many examples.
good
Great
This has been one of the best courses I've taken at UofT! I have loved the content of the course but also the instruction in this course has been incredible. Prof. Balazs is an excellent lecturer and his explanations in class are very helpful. His office hours as well are very informative. What I am most impressed by is the speed at which questions are answered on Piazza. More often than not, questions are answered within 1hr of them being posted which is record time for any Math/CS course I have taken at UofT to date. Course notes are comprehensive, Prof. Balazs is very approachable and kind, problem sets and the midterm have all been fair. Prof. Balazs might be the most friendly math instructor at UofT I have to date and for that, I thank him very much. The only criticism I have would be that at times Prof Balazs can be a bit disorganized in his lectures with what he writes on the chalkboard and explains. But even then, if anyone has a question, he is always more than willing to explain further which makes up for any disorganization.
Elek was very personable and provided a good, fun learning environment. He was, at times, quite disorganized however and seemed unprepared for the lecture which was frustrating.
I cannot understand this course well, especially meet new questions, maybe the reason is that the course is difficult. Tough life.....

8. Please comment on any assistance that was available to support your learning in this course.

Comments
maybe post problem set answer after due
Great ta, Monday 1 pm section
Piazza was extremely helpful, it should be mandatory in every course. I'd suggest anonymous questions even to Professors, because I feel worse knowing my Professors will read my questions and might think it's stupid.
Office hour really helps alot
I really love all the notes post on course webpages, it helped me a lot.
Very quick responses to questions
Piazza was a MAJOR help (besides office hours).
problem sets helps a lot for practice after class. TA is great and answer questions with patience,
good
I liked the tutorials but it would be better if the pace in tutorial is faster so we can actually finish answering them in class
Prepare more for the class, organize the teaching material more.
Problem sets were very helpful with understanding the course materials. There were lots of office hours and my professor even spent extra time to answer our questions.
Great
The instructor's collective use of Piazza in this course has been incredible. The amount of support for Piazza has been incredible! My TA has likewise been very approachable and always willing to help with any questions I may have had. The approachability and friendliness of the instructors of this course is also very very high and much appreciated. The course notes were also very helpful.
Elek's office hours were very helpful and he was extremely welcoming.
No assignments answers.

Part B. Divisional Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
FAS001 The instructor (Balazs Elek) generated enthusiasm for learning in the course.	4.4	5.0

Scale: 1 - Very Light 2 - Light 3 - Average 4 - Heavy 5 - Very Heavy

Question	Summary	
	Mean	Median
FAS002 Compared to other courses, the workload for this course was…	3.8	4.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - Strongly

Question	Summary	
	Mean	Median
FAS003 I would recommend this course to other students.	4.0	4.0

Part C: Departmental Items

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) explained concepts clearly.	3.9	4.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) was approachable.	4.4	5.0

Scale: 1 - Not At All 2 - Somewhat 3 - Moderately 4 - Mostly 5 - A Great Deal

Question	Summary	
	Mean	Median
The course instructor (Balazs Elek) answered questions clearly.	4.2	4.0

Scale: 1 - Poor 2 - Fair 3 - Good 4 - Very Good 5 - Excellent

Question	Summary	
	Mean	Median
UNIT(OQI) Overall, the quality of instruction provided by (Balazs Elek) in this course was:	4.1	4.0

Cornell University
Course Evaluation Response Summary
Semester: Spring 2018 Course Owner: MATH
Course: MATH 1110 LEC 001 CID: 5651
Title: Calculus I
Instructor: Elek
9 Responses, 13 Enrolled, 69.23% Response

Question	Mean	StDevP	Count	1	2	3	4	5	6
1. Did the lecturer stimulate your interest in the subject? [L1] 1 = not at all 5 = stimulated great interest; inspired independent effort	4.11	0.73	9	0	0	2	4	3	
2. Was the lecture presentation organized and clear? [L2] 1 = disorganized and unclear 5 = very organized and lucid	4.33	0.66	9	0	0	1	4	4	
3. Was the lecturer willing and available to help you overcome difficulties in this course? [L3] 1 = was of no help 5 = was very helpful	5.00	0	9	0	0	0	0	9	
4. Rate the overall teaching effectiveness of your lecturer compared to others at Cornell. [L4] 1 = worse than average 5 = much better than average	4.56	0.49	9	0	0	0	4	5	
5. Was the homework returned in a timely manner? [G1] 1 = never 5 = always	4.56	0.68	9	0	0	1	2	6	
6. Were the grader's comments helpful? [G2] 1 = no help 5 = very helpful	3.56	1.16	9	1	0	3	3	2	
7. How valuable were the homework assignments? [C1] 1 = taught me little 5 = extremely educational	3.89	1.09	9	1	0	0	6	2	
8. Rate the examinations in this course as a test of your knowledge. [C2] 1 = too easy, not adequate 3 = adequate 5 = too difficult, not a fair test	3.78	0.78	9	0	0	4	3	2	
9. Rate the level of difficulty of this course. [C3] 1 = too easy 5 = much too hard	4.25	0.96	8	0	1	0	3	4	
10. How suitable was the textbook? [C4] 1 = lousy 5 = great	3.78	0.78	9	0	0	4	3	2	
11. How many hours each week (on the average) did you spend on this course outside of class? [C5] 1 = 0-4 hours 2 = 5-7 hours 3 = 8-10 hours 4 = 11-13 hours 5 = 14 hours or more	2.89	1.28	9	2	1	3	2	1	
12. What was your most important reason for taking this course? [C6] (Use the answer that is closest to correct.) 1 = field or major requires it 2 = prerequisite for further courses 3 = interest in the subject matter 4 = reputation of the course 5 = reputation of the instructor 6 = distribution requirement	1.67	1.56	9	7	1	0	0	0	1
13. Have you attended homework study sessions this term? 1 = Yes, I attended often. 2 = Yes, I attended occasionally. 3 = Yes, I attended once or twice. 4 = No, I chose not to attend. (Please explain your reasons for not attending in the second comment space below.)	3.11	1.19	9	2	0	2	5		

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16. If you have any comments about the instructor, please include them here.

398. He was really a master at the subject material. Although lecture was disorganized and I could tell we had to rush through a few things he was very good at explaining complex topics. I really enjoyed class with my instructor.

25522. Explained things well, he made the class feel pretty low pressure which was nice

22032. Very helpful, nice, and encouraging.

8151. Very good and wasn't boring.

8165. Balasz was a great instructor, really glad I had him. Just a very nice and patient guy.

28909. Sometimes ran out of time to go over worksheets in class

11546. Balazs was a very good instructor. He always made sure that we understood the subject material and he always explained concepts in multiple different ways so that we'd understand them. He was also very nice and accessible during office hours

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[KB12] Manu Kapur and Katerine Bielaczyc. Designing for productive failure. *Journal of the Learning Sciences*, 21(1):45–83, 2012. 3

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